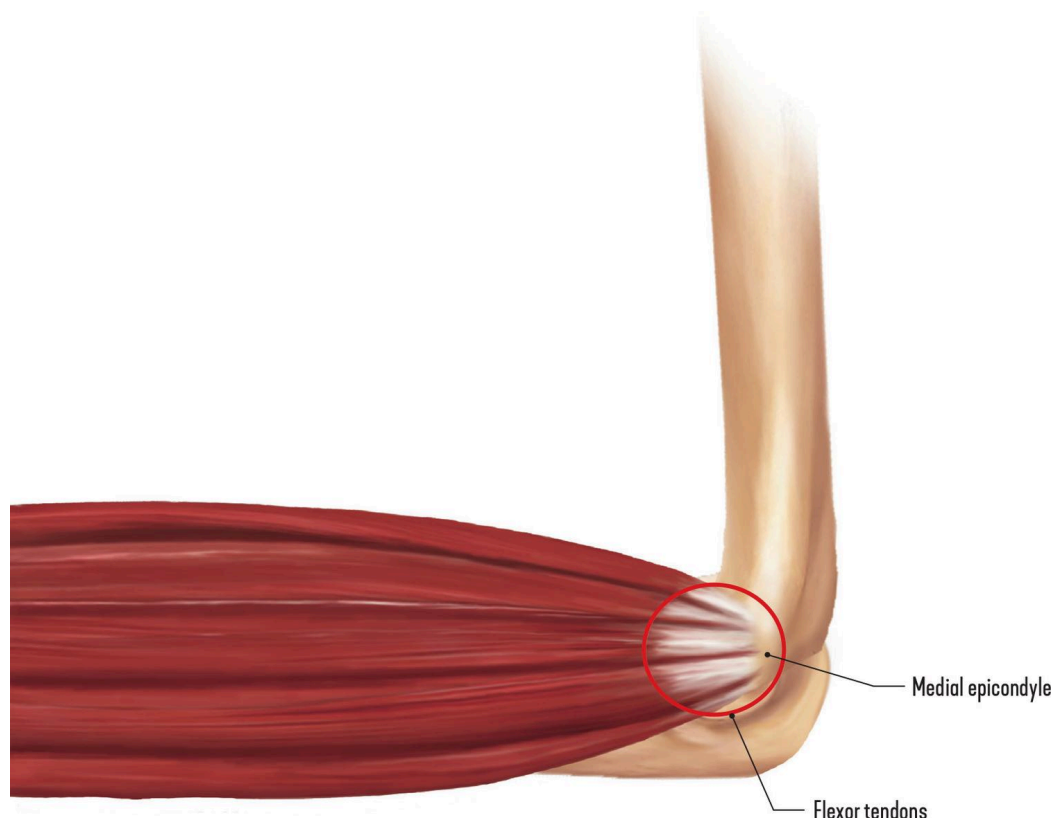


ELBOWS AND FOREARMS



Injuries and pain related to the elbows and forearms are mostly caused by excessive stress over time. However, the elbow is also prone to acute injuries from falls, especially during bouldering. Acute traumatic elbow injuries should always be evaluated by a healthcare professional to detect possible fractures and ligament damage; these injuries are not covered here. This section describes load-related issues related to the forearm muscles and their tendon origins on the inside and outside of the elbow, and pain in the middle aspect of the elbow.

MEDIAL EPICONDYLALGIA – AKA ‘GOLFER’S ELBOW’

‘It hurts when I press here, on the knuckle on the inside of my elbow. It’s painful when I warm up, but climbing can be fine. I notice that it gets worse after climbing, and then it hurts if I slice bread or twist bottle caps.’

The ‘knuckle’ referred to here is the bony protrusion on the inside – medial side – of the elbow. The knuckle is called the medial epicondyle, so medial epicondylalgia simply refers to pain in this area. The condition is also known as ‘golfer’s elbow’, although a quick Google search shows that it is now also referred to as ‘climber’s elbow’; however, since this name is reserved for another condition (see page [96](#)), we will stick with golfer’s elbow or medial epicondylalgia.

The common tendon for most of the muscles that flex the fingers and wrist and rotate the forearm so that the palm faces downwards (pronation), originates from the medial epicondyle, and the pain primarily relates to reactions in this tendon. The tendon can react due to excessive strain within a short period, but it is also subject to compression forces where it spreads out over the knuckle – particularly when we fully flex our elbows when locking off. Imagine what happens during a pull-up: we grip a bar with our palms facing away from us (a pronated position), using both the flexor muscles of the fingers and wrist and the pronator teres muscle while pulling up into a fully flexed elbow position. If we combine these forces of tension and compression, along with numerous repetitions, the tendon can react and cause pain. Similarly, a high climbing volume on steep or vertical terrain can cause excessive strain, leading to a negative reaction in the tendon.

Although the reaction is a result of doing too much, the solution is not to avoid loading the tendon but rather to break the load down into different components and train them separately.

It is important to emphasise that there is no perfect protocol or training method; instead, we must rely on some basic principles of loading and adaptation in tendon tissue. This provides us with ample opportunities to

customise load management in climbing, exercise selection and dosage according to what suits the individual concerned.

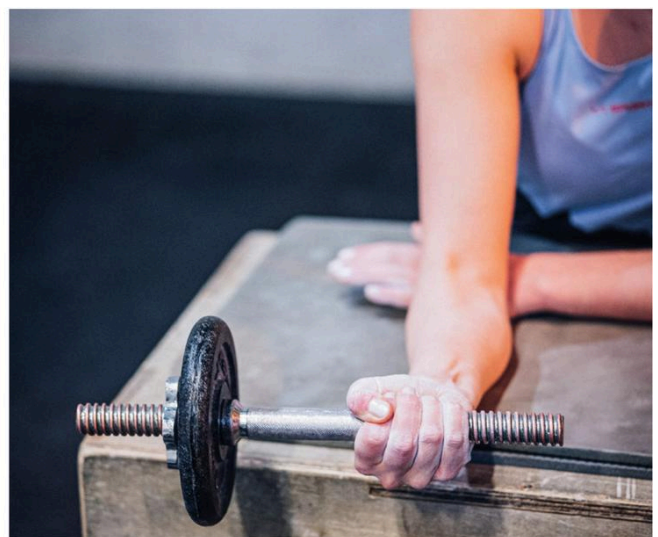
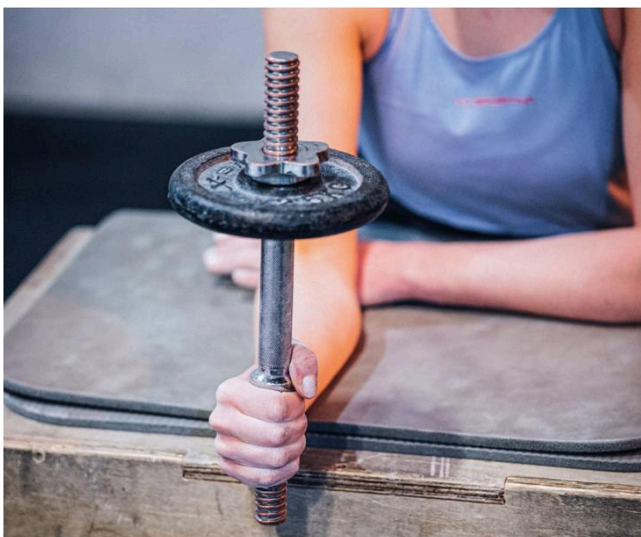
Once you experience pain on the insides of your elbows, it's natural to want to climb easier routes or boulders. Although this may seem rational, it is not necessarily the best strategy because most people will climb more when they climb easier. Therefore, the volume increases, and even if you don't climb as hard, the increased volume itself can greatly provoke the tendon. This particularly applies to climbing that is steeper than vertical, since the flexor muscles still have to work relatively hard. Therefore, if you want to climb easier stuff for a while, it is recommended to climb on less steep wall angles and focus on technical elements such as foot placements, balance and weight transfer.

However, it is not necessary to reduce climbing intensity – it is simply the volume that must be reduced. This means shorter sessions, fewer attempts and fewer moves per session, but there's nothing wrong with climbing hard and steep routes or boulders. I have already explained on the previous page why you should avoid deep lock-offs, and although you may find that individual moves can have a negative impact, you can then choose a different boulder or route. The most important thing is still to manage training load according to the so-called '24-hour response' – where the symptoms you experience after a session return to baseline within approximately 24 hours (see page [93](#)).

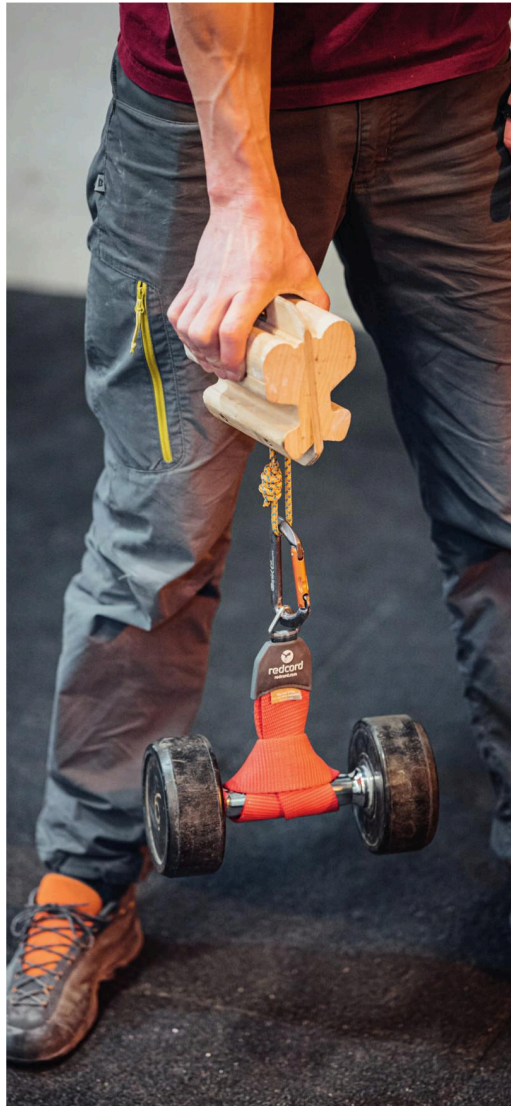
The principle for effectively managing tendon-related issues is to **load the tendon**, and it is crucial to recognise the possibilities this provides. There is a lot you can do even if your elbow is bothering you, and your goal should be to return as a better and stronger climber. You can achieve this by training technique, doing shorter sessions and working on both finger strength and overall upper-body strength – there is significantly more you can do than what you need to avoid. There are numerous examples of high-level climbers who have climbed hard routes and boulders despite dealing with golfer's elbow. They were able to accomplish this by being aware of which moves and types of climbing worsened their symptoms, so that they could choose different routes and boulders. This approach, combined with the fact that this condition *will* improve over time, is sufficient in most cases. However, there are exercises you can try if you want to be more specific in your approach.

Unfortunately, golfer's elbow tends to cause symptoms for several months and tends to flare up again when you exceed the limit of how much load the elbow can handle. It is important to be aware that this is a natural part of the process and you should use the symptoms as a tool to choose exercises, types of workouts and training methods. Through a combination of load management while climbing and doing specific strength training, you can effectively manage the condition until it eventually disappears.

EXERCISES



PRONATION AND SUPINATION. Using a dumb-bell with a weight disc at one end, start with the weight facing up and slowly lower it out to the side before returning to the starting position. Start with a weight that does not cause unacceptable pain and aim for 10–12 repetitions. Initially, you can do one such set every day, then increase the weight, reduce the number of repetitions, increase the number of sets per session and do only 1–2 such sessions a week. It may then look like this: 4–6 repetitions in 3–4 sets, 1–2 times a week.



PICK-UP WITH PINCH BLOCK. Start with a weight that you can hold for no more than 10 seconds. Repeat for 5 sets with 1–2 minutes' rest in between. Gradually increase the weight and reduce the holding time to 3–5 seconds per lift. Then do 4 lifts of 3–5 seconds with a 5-second rest in between and repeat for 4 sets.



ONE-ARM LOCK-OFF. You can either reduce your body weight or add extra weight to find the right load for you. Aim for a hang time of 10 seconds as a starting point and reduce the hang time to 5 seconds as you increase the weight. Repeat in 3–4 sets with 1–2 minutes' rest.



LOCK-OFF ON TWO ARMS and DEADHANG are good starting points for long hang times, i.e. hang times of up to 25–30 seconds. Isometric loading with a long time under tension means that forces within the tendon are reduced, allowing us to address the part of the tendon that is shielded. This shielding is called 'stress shielding' and is a mechanism to prevent further damage to the tendon tissue. You can start with a 15-second hang and gradually increase to 30 seconds and do 2 sets of lock-offs and 2 sets of deadhangs.



JUMP-START. At the opposite end of the scale from 'stress shielding' and 'stress relaxation' are training methods that stimulate stiffness in the tendon tissue and improve the tendon's ability to store and release elastic energy. An example of a training method for these properties is jump-starts, where the focus is on landing the top hold. You can do 4–8 such jumps per arm and repeat for 3 sets.

TENDINOPATHY

Changes in tendon tissue can occur as a natural part of the ageing process, or as a reaction to both excessive and insufficient load. These changes result in reduced tendon fibre quality and an ingrowth of blood vessels and nerve endings into the tendon tissue. However, such changes to the tendon are also seen in people without symptoms, and it is only when there are both tissue changes *and* pain during loading of the tendon that we call it *tendinopathy*. This term has replaced the previous term 'tendonitis', where 'itis' refers to an inflammation that we currently have difficulties identifying. Calling such tendon-related problems inflammations is incorrect, and measures aimed at reducing an inflammatory response – such as rest, cooling and anti-inflammatory medication – will not have the desired effect.

What is important for our purposes is that **tendons adapt to load by being loaded**. There are many ways to load tendon tissue, but since load is crucial for both resolving symptoms and increasing the capacity of the tendon to handle load, complete rest is rarely a good option. Australian tendon researcher Jill Cook has proposed a progression model for tendinopathies, which serves as a good framework for adapting exercises, load and activities. By following such a progression model, we can always adjust the load based on symptom response and gradually build up the capacity to tolerate whatever we expose our tendons to.

As an addition to this model, it should be noted that short sessions done frequently are suggested as a method to stimulate the tendon cells. As little as 10 minutes of low-load exercise stimulates the cells to produce more collagen tissue, which is beneficial for a tendinopathic tendon. After a session, it takes about six hours for the cells to become responsive to load again, so a protocol early in rehabilitation could be one short session three times a day with six hours' rest between each session.



NO-HANG. 10 seconds on/30 seconds off x10 in different grip positions. This will work all the muscle-tendon units involved in a grip position, including the tendons in the elbow.

One way to do this is through so-called *no-hangs*, where you can load for 10 seconds, rest for 30 seconds and do 10–20 repetitions.* The load should be low, preferably down to 30 per cent of 1RM (one repetition maximum), and the aim is not to tire the muscles.

Further training of the tendon should mainly focus on progressively heavier loads, since mechanical tension is what makes the tendon stronger and stiffer. A proposed protocol to stimulate this is 3 seconds' effort, followed by 3 seconds' rest, x4 repetitions, x5 sets, where the intensity in each repetition is over 90 per cent of 1RM. This can easily be done as isometric training on a fingerboard, using a grip tool or with weights depending on which tendons you primarily want to target.

To supplement this 'heavy loading' regime, you can expose the tendon to energy storage and release. This includes reducing the mechanical load but increasing the velocity and/or rate of force development. Following these principles, you'll build up the muscle-tendon unit to its previous capacity – and beyond.

* No-hang means, as the name suggests, that you are not hanging from the edges but standing on the ground and pressing your fingertips against the grip surface, whether it is a fingerboard or a portable grip tool.



STEP 1:

Isometric training without compression.



STEP 2:

Strength, without compression. Kinetic chain, functional for the activity.



STEP 3:

Energy storage. Faster movements. Eccentric in the outer position. Exercises at a faster pace.



STEP 4:

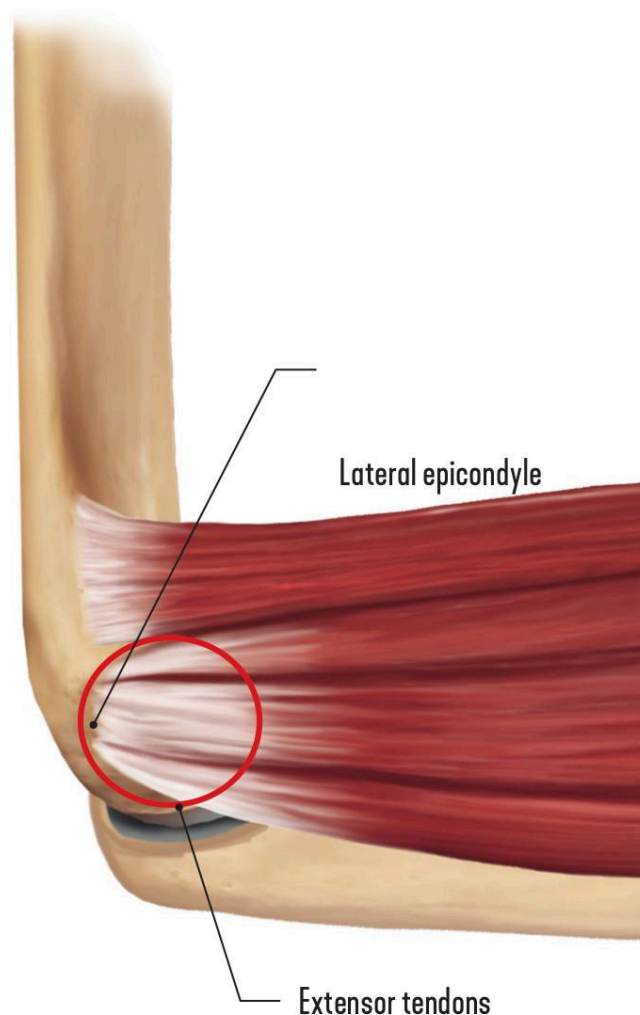
Energy storage and release, stretch-shorten-cycle (SSC). No restrictions, sport specific.

Both the no-hang and heavy load protocols can be used for different specific exercises, aiming to create mechanical tension which again stimulates adaptation in the tendon tissue. It's therefore possible to combine this training with more functional exercises, which will include far more elements than just adding mechanical tension to the tendon.

24-HOUR RESPONSE

It can be difficult to use pain as an indicator of tissue damage, or as a sign of overdoing things. As we will see later in the book in chapter 3, pain is an experience that is based on far more than tissue damage, yet it is still the symptom we're left with and which we must try to make sense of after exercise. Dealing with this symptom response from a 24-hour perspective makes us less vulnerable to transient changes in pain and allows us to look more towards a trend line that develops over time. Naturally, it can hurt to load a painful body part, and it is likely that a symptom response will be experienced after exercising throughout a rehabilitation process. However, as long as these symptoms return to baseline within 24 hours, we can continue rehabilitation and see that the trend line moves in the right direction – despite transient pain responses along the way.

LATERAL EPICONDYLALGIA – AKA ‘TENNIS ELBOW’



‘It hurts here, over this knuckle. I notice it especially when I take a milk carton from the fridge, or when I unscrew the lid of a jam jar. After climbing, it hurts even more, and my elbow feels stiff when I wake up in the morning.’

The description above is not unlike that of medial epicondylalgia, but this time it concerns the knuckle on the outside – lateral side – of the elbow. Therefore, the condition is called lateral epicondylalgia, but it is better known as ‘tennis elbow’. If you’re not familiar with tennis, hitting a ball with a racket involves gripping around the racket and angling your wrist backwards during the swing phase. This backward angling is called wrist extension, and the muscle-tendon units that perform this movement originate from the lateral epicondyle. Rather than hitting a ball with a racket, more relevant for us climbers is that we’re completely dependent on wrist extension for generating force when gripping. Just try flexing your wrist while loading your fingers in a half-crimp position – you won’t be able to generate much force. So, every time we hold on to a grip, we automatically extend our wrists and thus load our extensor muscles along with their associated tendons. We also rely on the extensor muscles in our fingers and wrists to maintain a grip position, and the importance of wrist extension is nicely illustrated through the phenomenon known as ‘chicken winging’, which occurs when we’re pumped.

Tennis elbow is a tendinopathy, just like golfer’s elbow, so the principles described on pages [92–93](#) for how we manage tendinopathies also apply in this case. By adjusting the load (primarily by reducing the volume), avoiding deep lock-offs, supplementing with the exercises on pages [89–91](#) and then gradually increasing the volume again, we will in most cases be able to train and climb throughout rehab. Despite tennis elbow – like golfer’s elbow – being able to cause symptoms for several months, there is still more we can do than we should avoid, and we can control the training load based on the 24-hour response described on page [93](#).

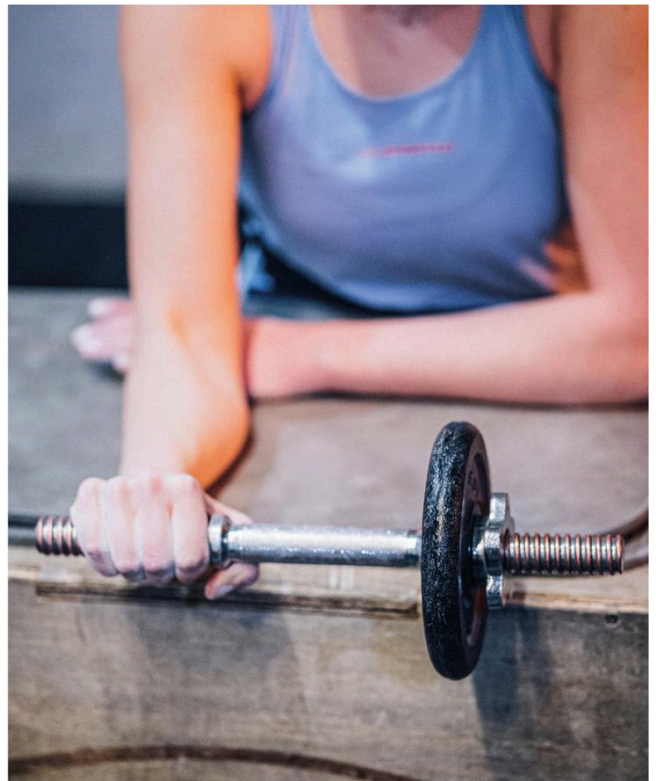
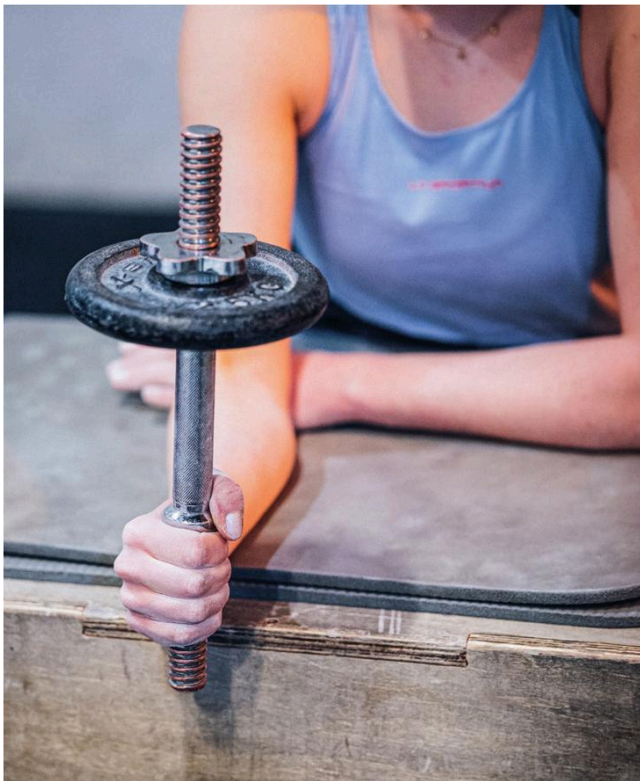
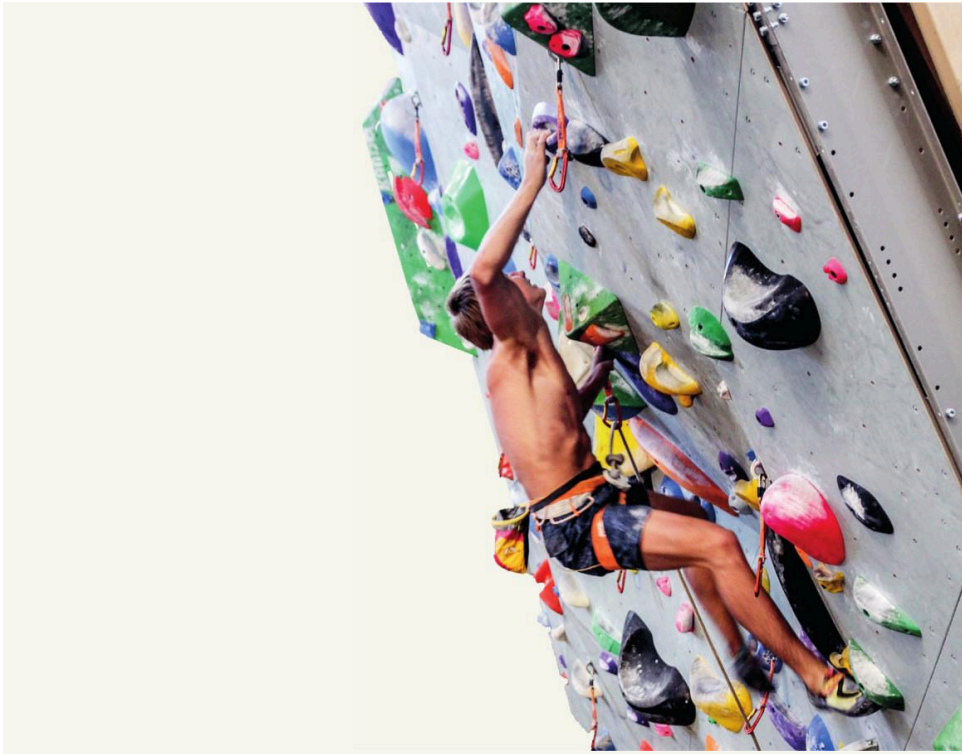
See exercises on pages [89–91](#).



LOAD CELL. Wrist extension provides a direct mechanical pull on the tendon insertion at the lateral epicondyle. By using an isometric training method with a load cell, you can control how much effort you can put in before the pain becomes unacceptable, and you can further control the training by increasing the length of time you hold the position.

CHICKEN WINGING

Have you ever wondered why we climbers raise our elbows when we're getting pumped? This strange phenomenon we call chicken winging is caused by us automatically trying to achieve a better joint angle for the flexor muscles of the fingers. As these muscles tire when climbing, the elbows start to rise and bend backward in relation to the wrists, and it looks like we have chicken wings.



PRONATION AND SUPINATION. Using a dumb-bell with a weight disc at one end, start with the weight pointing upwards and slowly lower it towards the middle before returning to the starting position. Start with a weight that does not cause unacceptable pain and aim for 10–12 repetitions. Initially you can do 1 set a day and then increase the weight, reduce the number of repetitions, increase the number of sets per session and do only 1–2 sets per week. It may then look like this: 4–6 repetitions in 3–4 sets, 1–2 times a week.

CLIMBER’S ELBOW

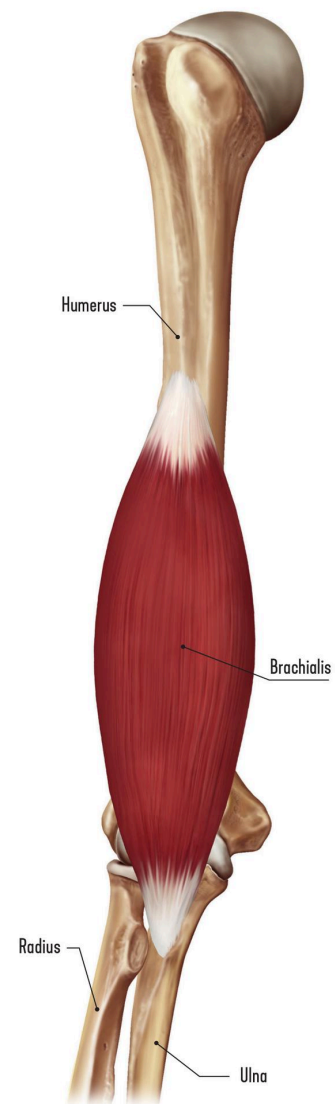
‘It aches here, around the biceps and down to the elbow. It feels sore and stiff and gets much worse after steep climbing or during pull-ups.’

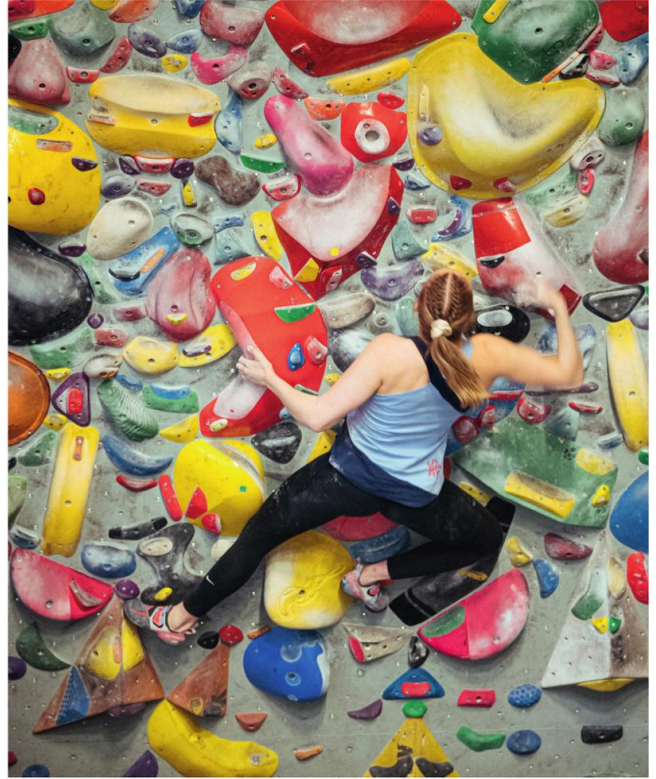
These symptoms, located on the front of the elbow, were described in the literature as far back as 1988 by Bollen, and were then called ‘climber’s elbow’. The symptoms were associated with pull-up training and traversing, and what these movements have in common is that the forearm is rotated so that the palm faces away from you while the elbow is flexed. With such pronation of the forearm, the biceps muscle is no longer the primary elbow flexor. This causes the brachialis muscle, which lies underneath the biceps and flexes the elbow regardless of the forearm position, to work harder.

Like both golfer’s elbow and tennis elbow, this condition can develop into a tendinopathy in the brachialis tendon, but this has not been investigated in research to date. Based on my own experiences with ultrasound scanning, however, it is extremely rare for me to see tendinopathic changes in the tendon itself; therefore, this condition appears to be more related to the muscle than to the tendon. The cause is almost always excessive and repetitive loading over time, including a lot of pull-up training and steep climbing – especially steep compression climbing on sloping holds where we ‘slap’ on to grips. In addition to the previously mentioned positioning of the forearm, climbing on steep wall angles causes us to work statically with flexed elbows and with rapid transitions between eccentric and concentric phases, where we respectively grip and move on with the opposite arm.

The solution is, of course, to reduce the strain on the brachialis muscles by training at less steep wall angles and stopping pull-up training for a period. This does not mean that you cannot train strength, but that you must adjust the forearm position to neutral or supinated. You can therefore train one-arm lock-offs, with or without assistance, biceps curls and pull-ups with neutral or supinated grip. You can also adjust the wall angle slightly, for example down to 30–40 degrees, and climb on smaller holds with shorter moves; this will put less strain on the brachialis muscles than large moves on bigger holds on steeper terrain.

As with tendinopathies, loading the brachialis muscles is essential in order to return to climbing. Therefore, it is likely that there will be some pain and discomfort associated with training, but as long as you are able to perform the exercises without pain hindering force development, and there is no persistent pain response afterwards, the discomfort during and after training is completely okay. This is a harmless condition that in most cases improves completely just by managing the load and avoiding the worst triggers for a short period of time.





COMPRESSION CLIMBING ON SLOPERS. The body position is maintained by compressing between the holds, i.e. squeezing your arms together. In one movement, you release this tension when you let go with one hand and regain it as quickly as possible when you reach the next hold. This combination of rapid force development, eccentric loading in the landing phase and a long period of brachialis tension is a likely cause of anterior elbow pain.

EXERCISES



BICEPS CURLS. Place your arm over a bench and start with the elbow straight. Slowly raise the arm and return it to the starting position. In this way, you can work the brachialis muscle hard while it is supported by the biceps muscle. Aim to work as hard as you can for 4–6 repetitions in 2 sets.



ASSISTED ONE-ARM LOCK-OFFS. Do 2 sets of 10 seconds each, at both 90 degrees and 135 degrees of elbow flexion.



UNDERHAND PULL-UPS (AKA CHIN-UPS). Start with your arms straight and pull up until your chin is over the bar. As with biceps curls, the biceps muscles now assist the brachialis muscles flexing the elbows. Either use bungee cords under your legs for assistance or add extra weight if necessary to complete 4–6 repetitions in 2 sets.

FUN FACT

The loading pattern with pronated forearm and repetitive load is commonly seen when bouldering on the sandstone boulders of Fontainebleau in France. Most of the boulders you will find in Font have a rounded top, often with nothing more than pure friction as the only means of topping out. You must then press the palms of your hands down on the holds at the same time as flexing your elbows to pull your body up. This movement pattern is central to almost all of the top-outs, independent of the level of difficulty of the boulder problem, and is repeated every day when you are out climbing in Font. Many climbers end up suffering from climber's elbow when visiting Fontainebleau, hence the nickname 'Font elbow'.



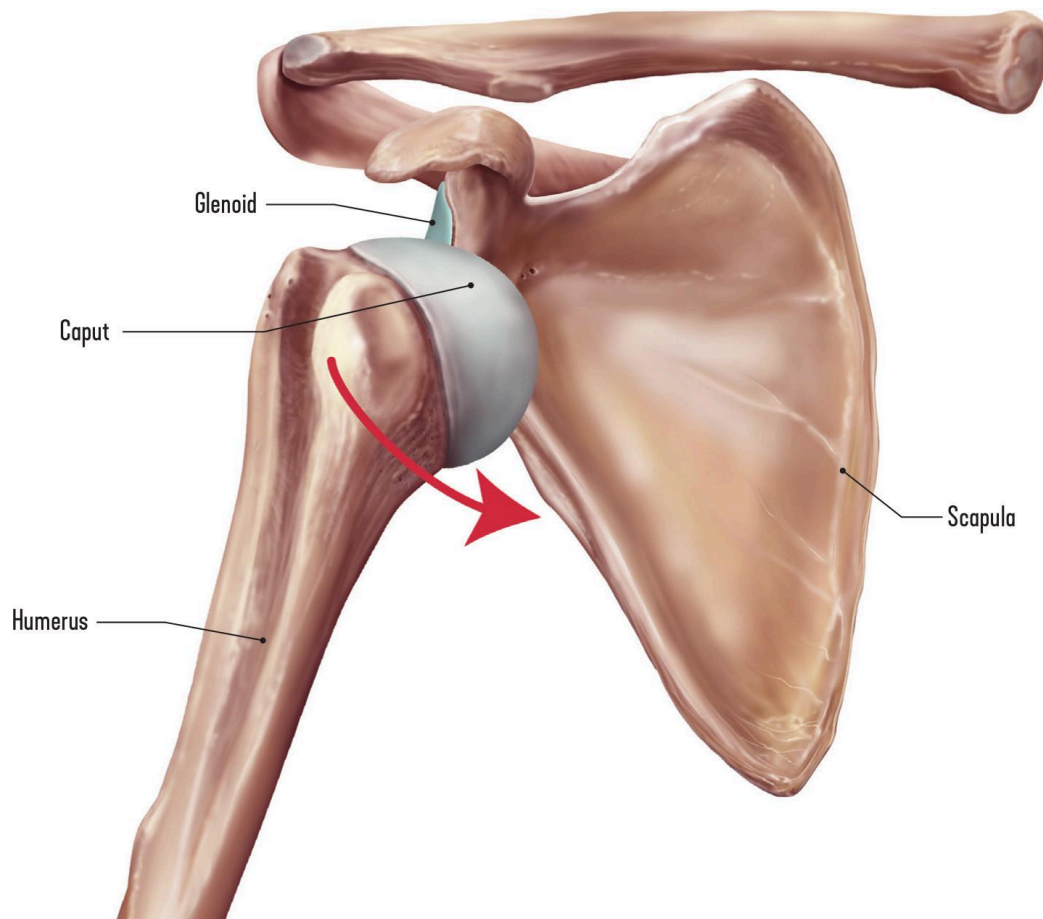
Two happy boys at the top of a boulder in Fontainebleau.

SHOULDERS

After the fingers, the shoulders are the next most common site of injury for climbers. However, unlike in the fingers, there are far more structures in and around the shoulder that are susceptible to injury, and it is beyond the scope of this book to describe all of these in detail. Instead, I will describe the most common climbing-related injuries and look at how we should handle them as effectively as possible.



The author in a double shoulder press on the first ascent of *Reisen til Ixtlan* (F9a), Hvarnes, Norway.

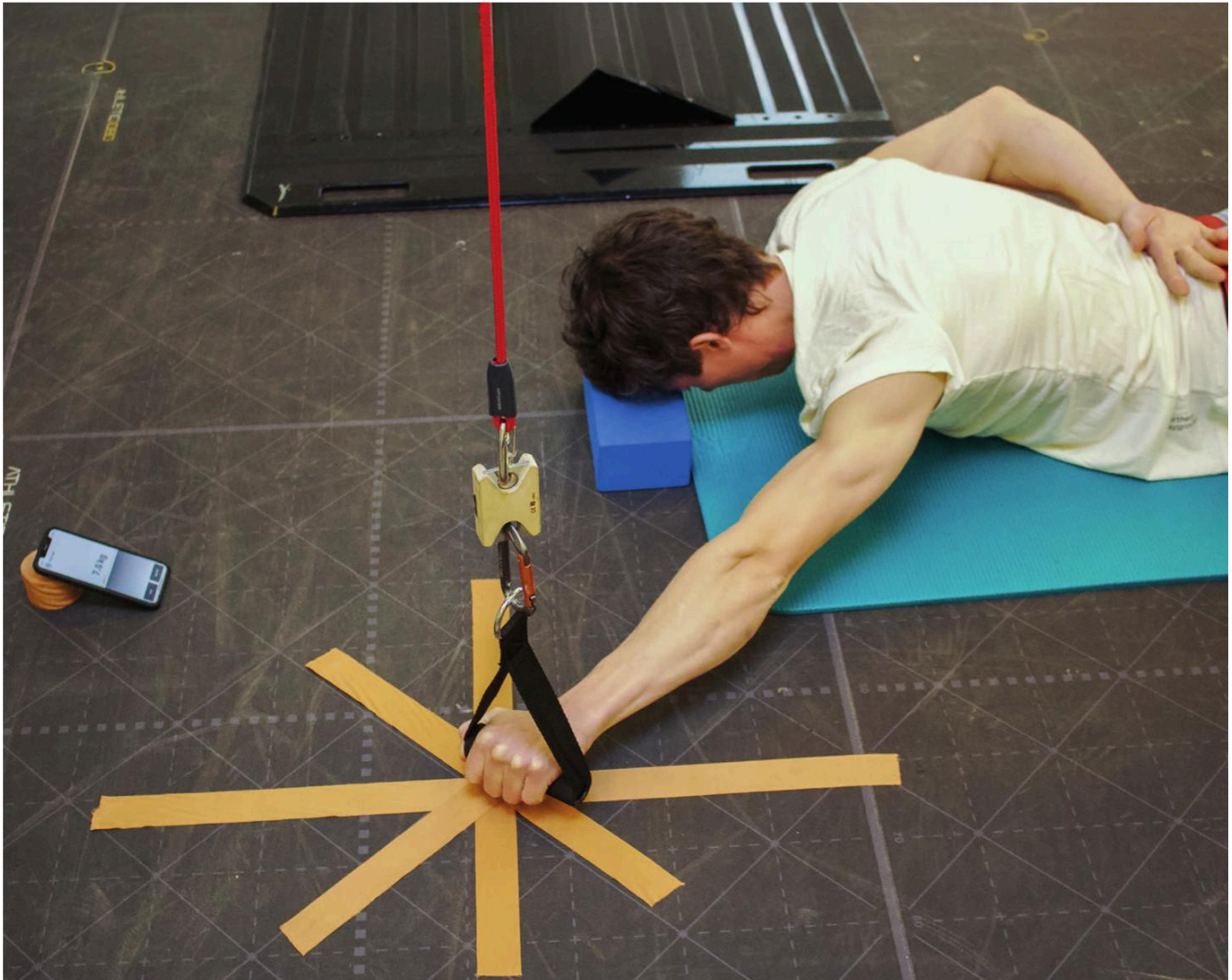


ANTERIOR SHOULDER DISLOCATION

If the shoulder dislocates as a result of an acute event, such as a fall or a heinous move, it will dislocate forwards in over 90 per cent of cases.

This dislocation involves the head of the upper arm bone sliding out of its socket in the shoulder blade. In some cases, it happens as a subluxation (partial dislocation), where immediately after dislocation it slips back into place on its own; however, it will usually remain dislocated and must be put back into place. Once this is done, an X-ray image should be taken to rule out fractures and ensure that the shoulder joint is properly positioned.

Since dislocating the shoulder involves structural damage to the joint, an MRI examination is also recommended to assess the labrum, the joint capsule and the rotator cuff tendons. This is important for assessing the stabilising structures, because the more structural damage there is, the greater the likelihood of future dislocations. And despite a thorough rehabilitation programme, unfortunately the chance of a subsequent dislocation is high, especially in younger and active people. Therefore, the situation should be discussed with an orthopaedic surgeon regarding whether surgical stabilisation should be considered or if the focus should be on rehabilitation and a return to climbing instead.



The ASH test is one of many options for the monitoring of shoulder and glenohumeral strength development during rehabilitation following an anterior shoulder dislocation.

Repeated dislocations cause more damage to the stabilising structures in and around the shoulder joint, thus further increasing the risk of recurring dislocations. These are important considerations to take into account when deciding whether or not to consider surgery.

Initially, the pain will limit how much the arm can be used, but it is still appropriate to start training from an early stage to stimulate the muscles without risking another shoulder dislocation.

The exercise selection can be done both with and without occlusion (see [box-out](#)), with the goal of preserving muscle mass and creating movement of the shoulder joint. In consultation with healthcare professionals, progression should continue towards heavier loads, increased range of motion and faster movements. The focus must always be on putting yourself in the best possible position to return to climbing, so the training should be both general *and* sport specific.

OCCLUSION TRAINING

This is also known as blood flow resistance training (BFRT), and the principle is to reduce the venous blood flow from the muscles.

Occlusion of blood flow leads to reduced oxygen supply and an accumulation of waste products in the muscles, thus increasing the metabolic stress, which is what we aim for when working a muscle to fatigue.

The combination of mechanical load and metabolic stress results in muscle growth, or at least a reduction of muscle loss during a period when we cannot load the muscles as before. The advantage of using straps to restrict circulation is that we can get away with both lower load and lower volume in an exercise, and yet still stimulate the muscles.

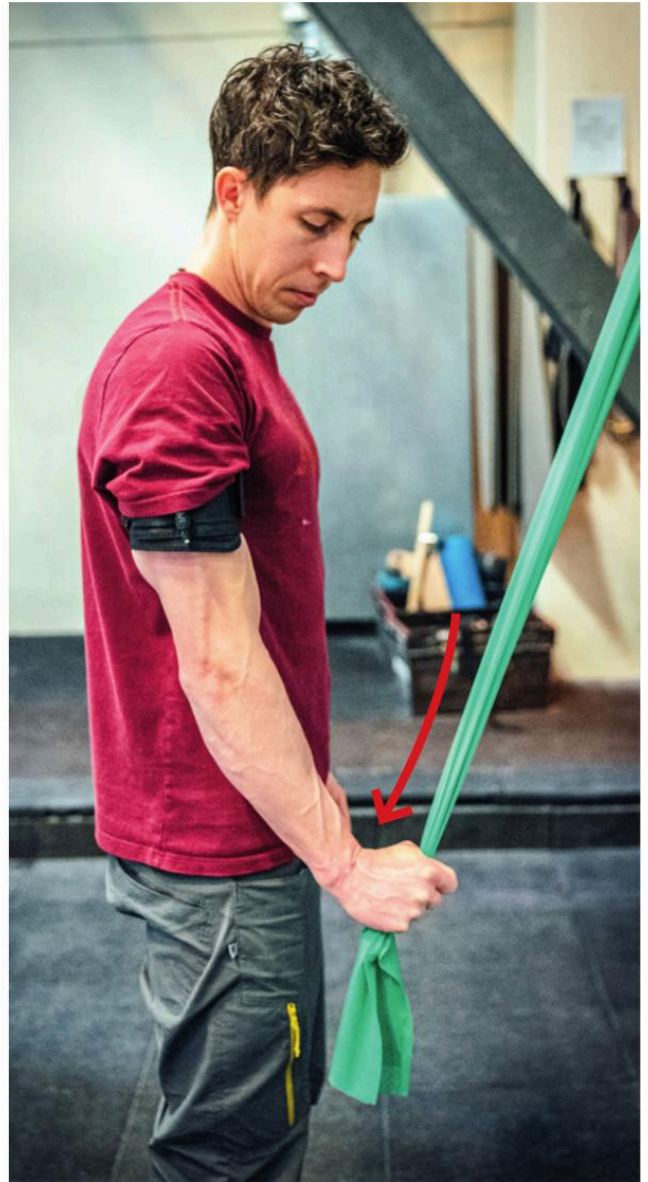
The exercise protocols presented here can be done with 1 set of 30 repetitions followed by 3 sets of 15 repetitions. Start with exercise 1 and work through all exercises before starting the next set. It should be uncomfortable but not painful, and there has been no evidence showing better training effect by increasing pressure.

The goal is to work with a constant pumping sensation without pain or tingling/stinging in the arms.

EARLY PHASE EXERCISES



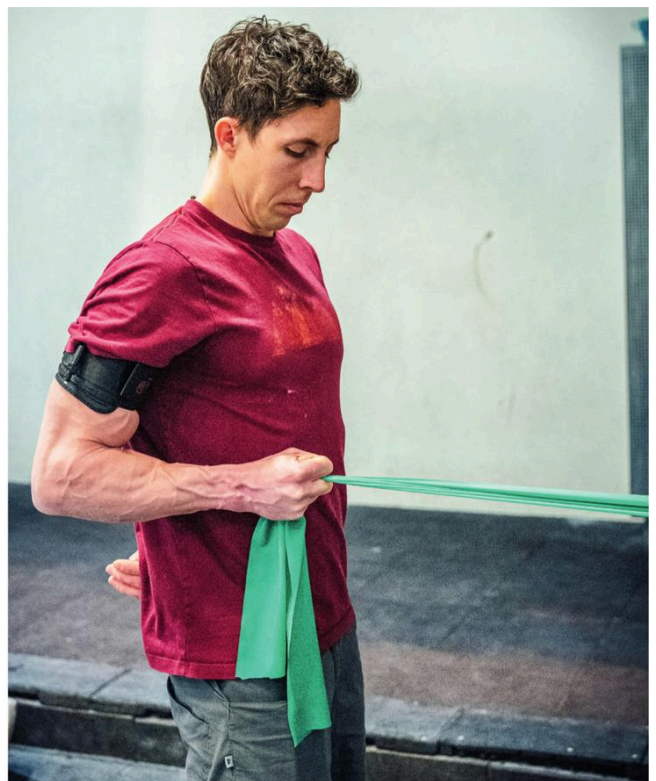
EXERCISE 1: FINGER CURLS. Using a light weight or resistance band, curl your fingers from an open position to a closed curl.



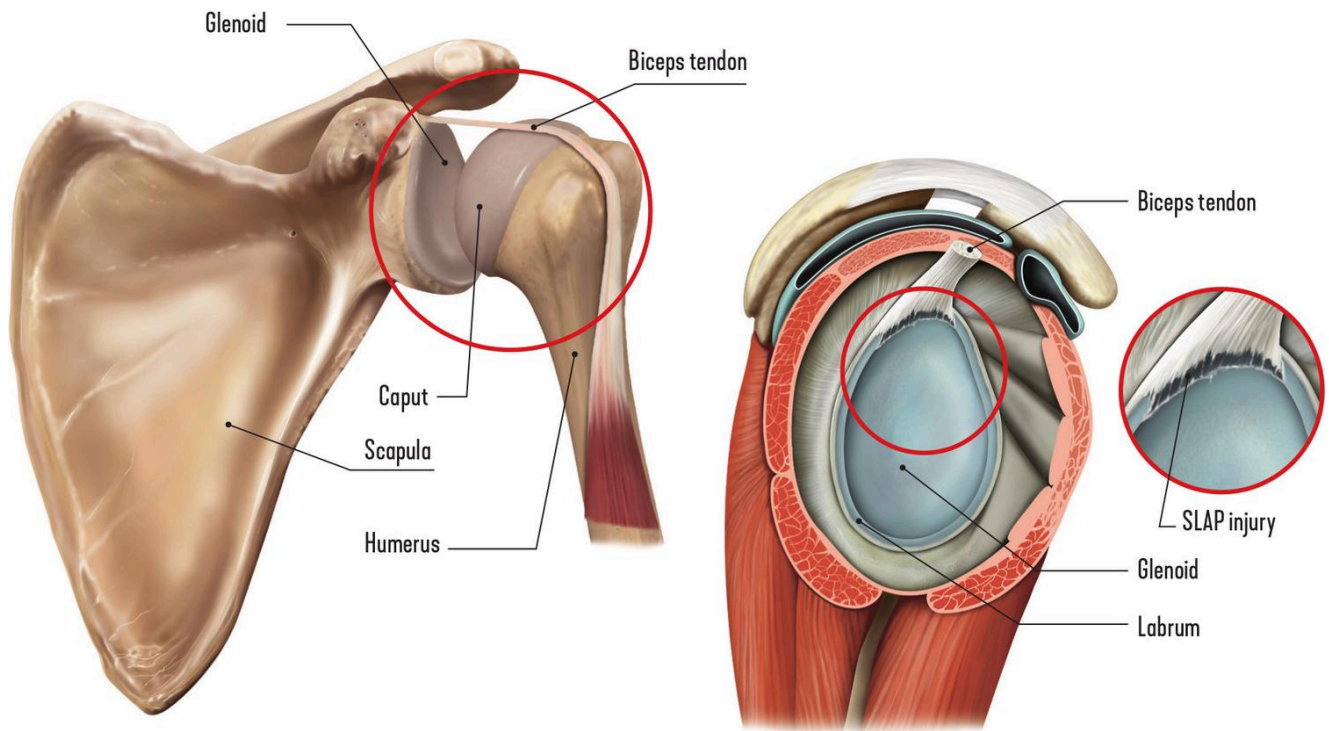
EXERCISES 2 AND 3: BICEPS CURLS and TRICEPS PRESS.



EXERCISE 4: PALLOF PRESS. Start with your arm at your side and your elbow at a 90-degree angle. Press your arm straight forwards without it moving in front of your body.



EXERCISE 5: ROWING. Starting with a straight arm, pull straight backwards, at the same time pulling your shoulder blade back and towards the centre.



BICEPS AND LABRUM

Around the glenoid cavity of the scapula, we have a labrum that functions as a suction cup on the joint head. This labrum allows the joint head to move around in the glenoid cavity without dislocating. At the upper aspect of this labrum, the tendon from the long head of the biceps attaches, and this site of the labrum can be damaged due to both acute trauma and microtraumas over time. When this injury is seen on MRI images, it is called SLAP – *superior labral anterior to posterior* – where the acronym describes an injury to the upper labrum that extends from the anterior (front) part to the posterior (back) part of the glenoid.

The mechanism of injury involves stress that causes the biceps tendon to pull the labrum off the bone, which can result in pain, clicking in the joint and a feeling of instability. However, such injuries, and changes in the biceps-labrum complex, are regularly observed on MRI images of athletes without any shoulder symptoms, so medical history becomes crucial for further management.

If you have had an acute incident and felt something happen inside your shoulder – for example, while taking a swing on a straight arm, or during a heinous shoulder press – and then experienced subsequent pain and a sense of instability, followed by an MRI showing a SLAP tear, these findings can be correlated. Since the labrum will not reattach itself to the glenoid, surgical stabilisation may be considered if systematic rehabilitation does not fully resolve the condition. Without a triggering traumatic event, there is much uncertainty about whether surgery would be beneficial; however, an individualised and well-designed strength and conditioning programme can allow most climbers to return to their desired level without undergoing surgery to the biceps-labrum complex.

EXERCISES



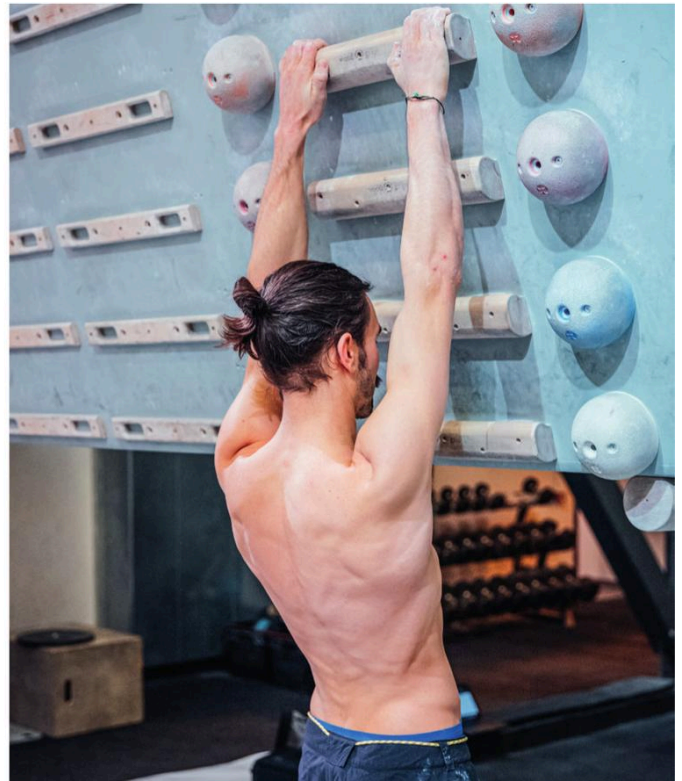
PUSH-UP WITH ROTATION. Do a push-up, then put all your weight on one arm and rotate your torso so that your free arm is pointing up towards the ceiling; repeat with the other side. Do 6–10 repetitions and 3 sets.



PULL-OVER. Lie on your back and hold a weight disc in one hand. Bring your arm straight back and down towards the floor. Make sure to keep your back in contact with the floor so that it doesn't arch upwards to help you lower your arm. Note that the bottom position is the most demanding, so adjust the weight so that you have control in this part of the movement. Do 6–8 repetitions and 3 sets as part of your warm-up for climbing, or as part of a strength training programme.



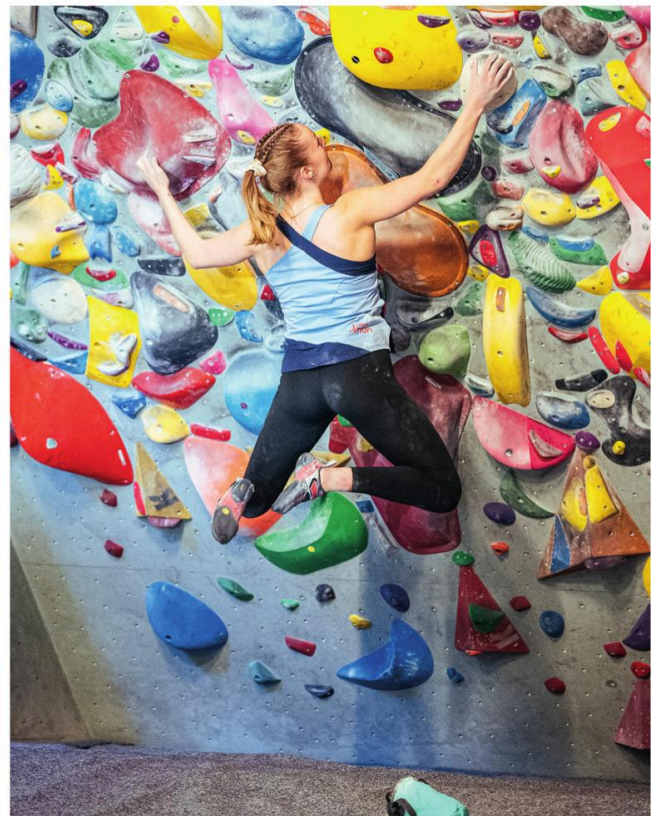
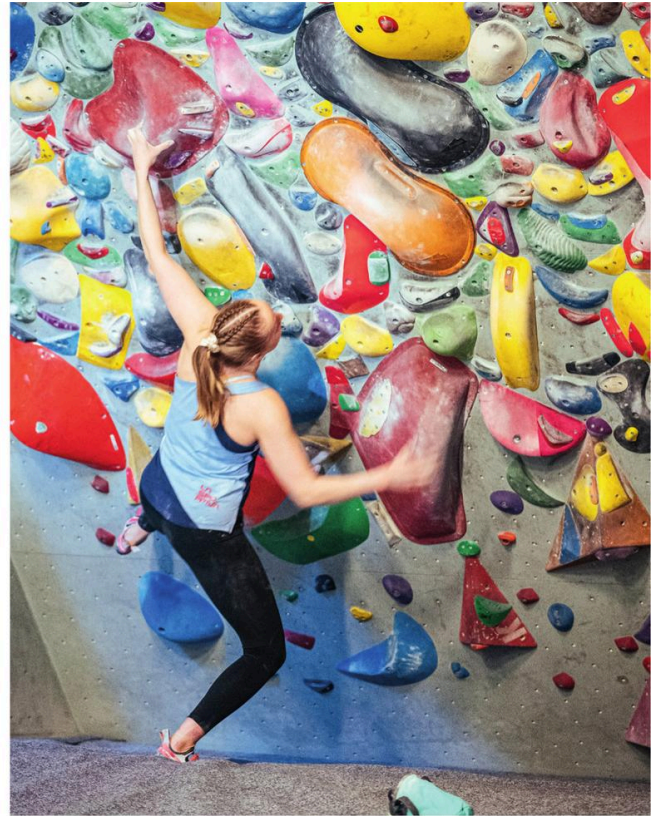
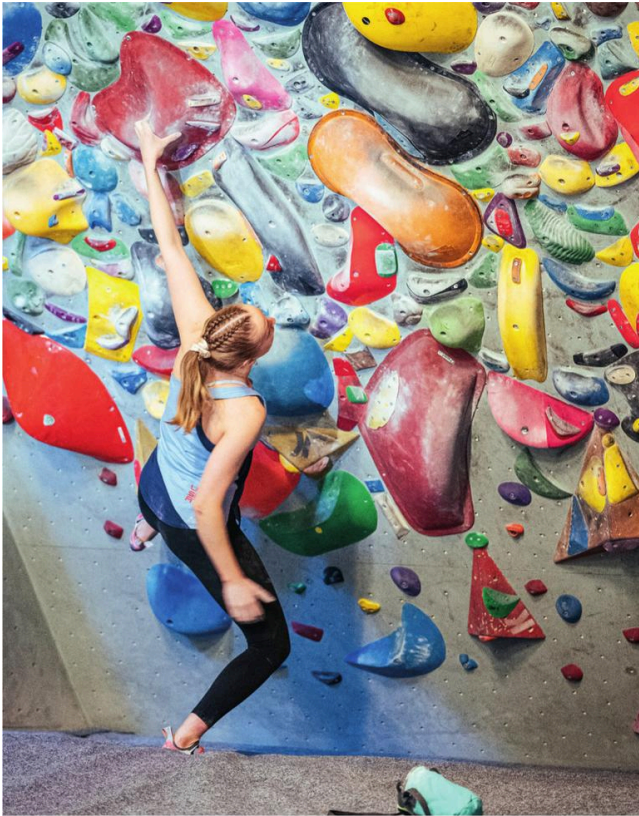
SUPPORTED EXTERNAL ROTATION. Holding a dumb-bell, let your arm rest on the inside of your knee with a 90-degree angle in the elbow joint. Bring your arm up so that the weight is pointing towards the ceiling and then slowly lower it back down. Repeat for 6–8 repetitions and 3 sets.



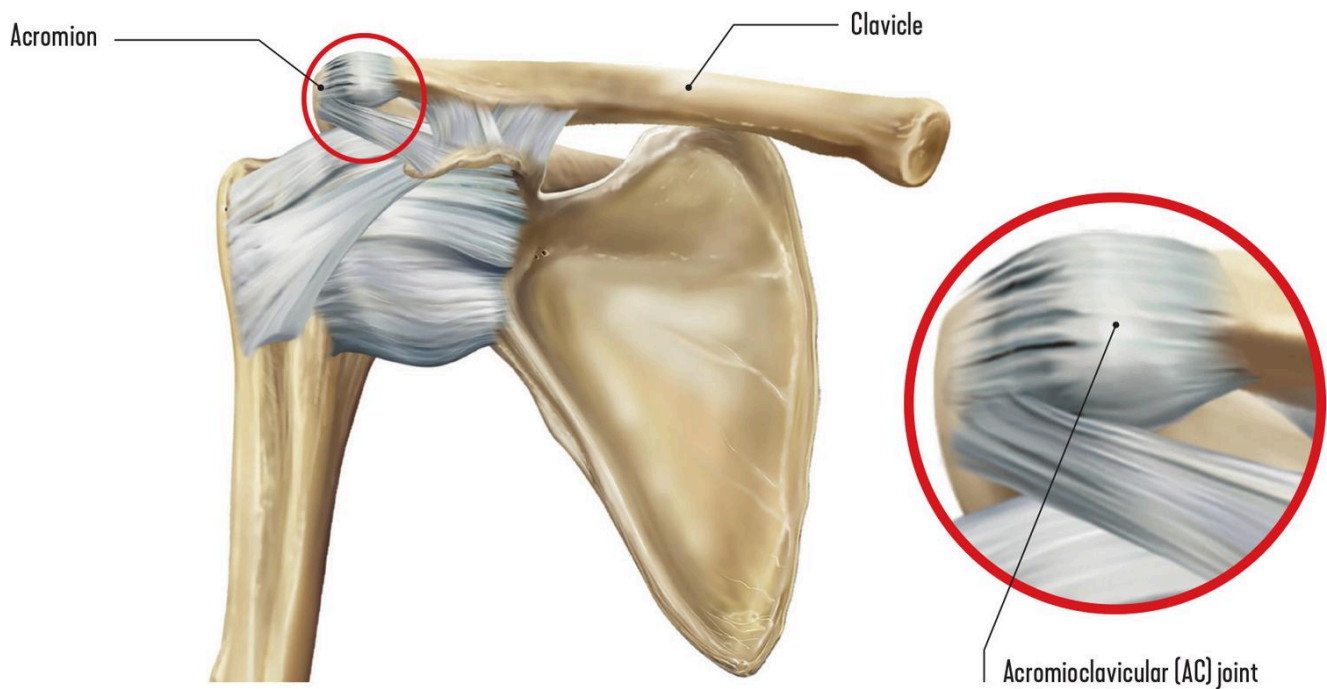
SCAPULAR PULL-UP. Hang on straight arms and pull your shoulder blades down and back so that you end up in a higher position but with your arms still straight. Slowly lower back down and repeat for 6–10 repetitions in 2 sets as part of your climbing warm-up (see page [34](#)).



ONE-ARM LOCK-OFF IN OPEN POSITION. With an open position (wider than 90 degrees of elbow flexion), the upper arm is farther away from the body than when locking off in a higher position, placing greater demands on the adduction muscles around the shoulder joint. Hold this position for 10 seconds until you are close to exhaustion at the end of the hang; repeat for 3 sets. I recommend gradually reducing the hang time to 3–5 seconds and increasing the weight.



JUMP-START. On a spray wall, you can practise jump-starts with different holds and varying distances between them. Large distances between holds require more strength and stability, so start with a short distance and increase the distance as you get stronger and more confident.



AC JOINT

The joint between the shoulder blade and collarbone is called the acromioclavicular joint, also known as the AC joint. The AC joint can be dislocated in a fall, and these injuries must be examined to determine their severity. The majority of such injuries are treated without surgery, and rehabilitation is guided by pain and range of motion; the prognosis is good.

For climbers, load-related AC joint pain is more common than fall-related pain. It can occur spontaneously, for example following a demanding shoulder move, or over time as a result of hanging from straight arms. The pain is localised at the top of the shoulder and is caused by irritation or inflammation within the joint. Like with the finger joints, excessive load affects the AC joint when the articular surfaces are compressed against one other. The key to resolving the problem is managing load and avoiding positions known to compress the joint, such as when the arm is fully elevated overhead or when the upper arm crosses the midline, such as during a shoulder press. As long as you're aware of these positions, you can train and climb around them. Symptoms often worsen if you try to work through these positions despite experiencing pain, but they will typically disappear on their own when you work around them.

More pronounced changes in the joint – i.e. osteoarthritis – may predispose individuals to recurring problems within their AC joints. However, many people diagnosed with arthritis based on X-rays or MRIs do not experience any symptoms from their joints; therefore, any imaging findings must be considered alongside a clinical examination and whether pain is present or not. For cases involving AC joint arthritis with recurrent inflammation and pain, measures such as cortisone injections and eventually surgery should be discussed with healthcare professionals.

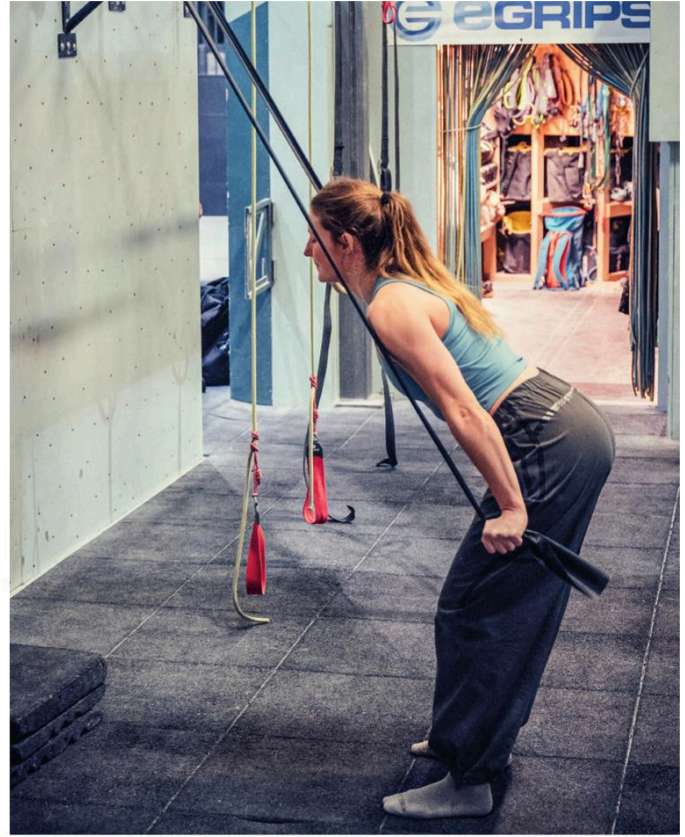


If the pain above the AC joint increases the higher you lift your arm, it is called 'high-arc pain'. This is a characteristic of AC joint-related pain.



CROSS-BODY TEST. Bring your arm in front of your body and lightly press the arm across the midline of the body. Pain above the AC joint is a sign of AC joint-related ailments.

EXERCISES IN PROGRESSIVE ORDER:



PULL-DOWNS WITH RESISTANCE BAND. Start with your arms at a height that does not provoke symptoms and pull the band down to your hips. This can be done as a warm-up exercise with about 20 repetitions and 2 sets.



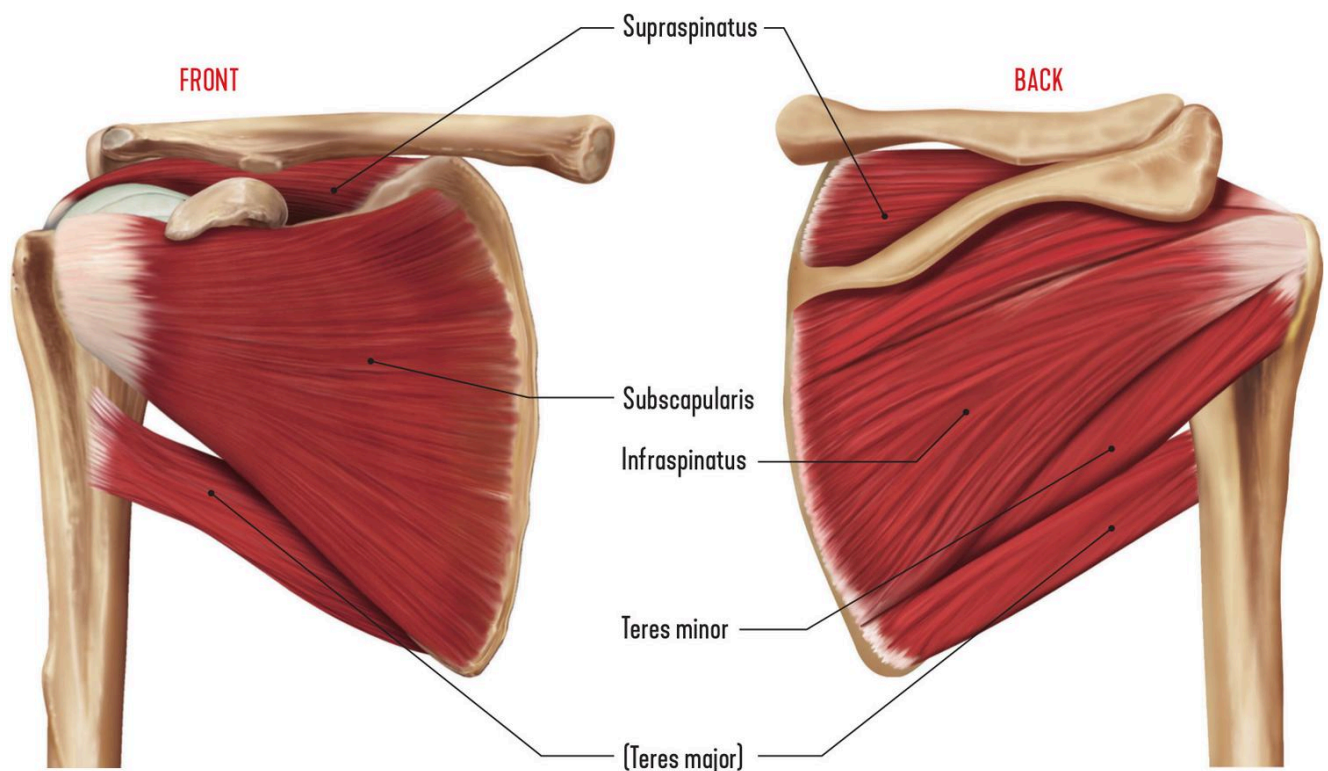
ONE-ARM ACTIVE HANG. Find the top position in a scapular pull-up and then hang by only one arm. This can be done with a band for your foot to relieve some weight if it is too heavy with your full body weight. Hold for 5 seconds and repeat for 5 hangs with 30 seconds' rest between hangs.



MID-RANGE PULL-UP. It's usually the bottom position, and in some cases the top position, that is troublesome. So, the solution for training strength can be to train in the so-called 'mid-range', i.e. avoiding the outer positions, but training heavily in the part of the movement path that is symptom-free.



PALLOF PRESS. Start with your arm at your side and your elbow at a 90-degree angle. Press your arm straight forwards, without allowing it to move in front of your body. Do 6–8 repetitions and 3 sets.



THE ROTATOR CUFF

The rotator cuff is a collection of four muscles – supraspinatus, subscapularis, infraspinatus and teres minor – with their associated tendons that attach as a cuff around the head of the upper arm. The shoulder joint can be compared to a ball (the head) resting in a socket (the glenoid), and it is this relationship between the ball and

the socket that makes the shoulder the most mobile joint in the body. However, mobility comes at the expense of stability, and the main job of the rotator cuff is to centre the ball in the socket when we move the arm.

As a result of traumas such as falls or other acute incidents, the rotator cuff tendons can rupture, and large cuff tears involving multiple tendons should be treated by an orthopaedic surgeon. The clinical signs of such injuries are an acute event followed by loss of strength and function of the arm. This strength loss is called a *lag sign*, and includes difficulties lifting the arm sideways, rotating the arm outwards or putting the arm behind the back. While pain may make it difficult to develop strength, lag signs are indicative of a tendon rupture where the muscle is no longer attached to the bone; reduced strength due to pain is another phenomenon, one that is also much more common than loss of strength due to extensive tendon tears.

Minor and partial tears of one or more rotator cuff tendons are relatively common. Like larger tears, smaller ones can occur acutely, but, generally, when examining people without shoulder pain or any previous shoulder injury, small tears are not uncommon in people aged 30 and older. Changes in tendon tissue are thus understood as processes that occur over time, and, similar to the elbow tendinopathies described earlier (pages [86–95](#)), they do not necessarily cause pain. Therefore, these small, partial rotator cuff tendon ruptures fall under the tendinopathy concept (pages [92–93](#)) when they are considered as the cause of symptoms – and they are managed accordingly.



LAG SIGN is the inability to perform a movement or maintain a position. Here we test abduction (photo 1), internal rotation (photo 2) and external rotation (photo 3 (Hornblower)). The inability to perform the movement, or the fact that it only takes the force of one finger to break the position, indicates greater damage to the rotator cuff. Consequently, you should be referred for further investigation.

The term *rotator cuff tendinopathy* is therefore proposed as a way to understand non-traumatic shoulder problems, and when these small tears are seen as part of the tendinopathy it means that tendon-related issues can occur even without ruptures. A symptom of tendon-related problems is pain on loading. The strength is there, but it hurts to use the arm. Through a review of someone's medical history and a clinical examination, we can identify probable triggers – which positions and movements increase symptoms – and then develop a plan for how to best manage the condition.

It might come as a surprise to learn that this plan will not include the classic rotational exercises with bands that have been recommended for years. The rotator cuff has multiple jobs, but, as noted earlier, its main job is to stabilise the ball in the socket when we move the arm. So, in order to train the rotator cuff effectively, we must therefore expose it to demands that increase its capacity to tolerate the load, movements and arm positions it will be exposed to when we go climbing. A combination of varied and sport-specific strength training is thus considered the most appropriate approach, and if we can manage this while simultaneously addressing the triggers, most people will experience relief from their symptoms and be able to return to their desired climbing level.

EXERCISES



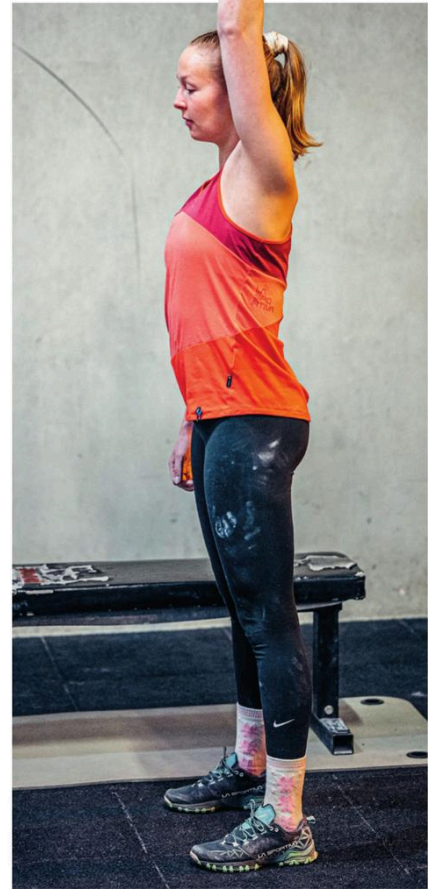
PALLOF PRESS. Start with your arm at your side and your elbow at a 90-degree angle. Push your arm straight forwards without allowing it to move in front of your body. This will work the upper and rear parts of the rotator cuff (supraspinatus and infraspinatus). Do 6–8 repetitions and 3 sets.



PULLING EXERCISES such as LOCK-OFF and JUMP-START train the entire shoulder complex, and particularly the front of the rotator cuff (subscapularis). They are climbing-specific and you can train with a high rate of force development, which is an important component in the final part of rehab before returning to sport.



ASSISTED EXTERNAL ROTATION. Holding a dumb-bell, let your arm rest on the inside of your knee with a 90-degree angle at the elbow. Raise your arm so that the weight is pointing towards the ceiling and then slowly lower it back down. Do 6–8 repetitions and 3 sets. This will work the upper and rear parts of the rotator cuff (supraspinatus and infraspinatus).



FLEXION RAISE. Starting with your arm by your side and keeping your elbow straight, raise your arm straight out and up above your head. Perform 8–10 repetitions and 2–3 sets. This will work the upper and rear parts of the rotator cuff (supraspinatus and infraspinatus).



BENCH PRESS WITH DUMB-BELLS. Lie on your back and push the weights straight up towards the ceiling. Slowly lower back down and do 6–8 repetitions and 3 sets to work the upper and rear parts of the rotator cuff (supraspinatus and infraspinatus).

A SHORT AND NERDY SHOULDER STORY

In 1972, orthopaedic surgeon Charles S. Neer introduced the term ‘impingement’ in reference to shoulder pain. Neer posited that the cause of shoulder pain was the ‘pinching’ of the rotator cuff tendons and the subacromial bursa underneath the acromion when the arm was raised. This hypothesis was tested by surgically removing the bursa and shaving off the underside of the acromion to increase the subacromial space. As almost all patients became symptom free after this intervention, ‘subacromial impingement’ became both a diagnosis and an explanation for its cause, and the term is still very much alive to this day. This is despite numerous high-quality studies which have shown that this procedure, called subacromial decompression, has no better effect than exercise or sham (placebo) surgery – or indeed that the condition simply resolves itself over time – for people with subacromial-related shoulder pain. What was once one of the world’s most common surgical procedures, undoubtedly helping many people, is no longer nearly as common, nor is it any longer recommended for managing shoulder pain.

WHY IS THIS STORY IMPORTANT?

If it were true that every time we lifted our arm above shoulder height, a bone spur drilled into both our bursa and rotator cuff causing shoulder pain, it would be highly unlikely that exercise would help. It would also be unlikely that symptoms would resolve on their own, and everything would point towards the need for surgical intervention to cure the condition. Such an explanatory model would therefore negatively affect our expectations towards both rehabilitation and prognosis – something we don’t want. So, not only has the impingement hypothesis been thoroughly disproven as an explanation for its cause, it also puts non-surgical approaches in a very poor light.

There are several other hypotheses and theories related to subacromial shoulder pain, and what they all have in common is regarding the use, movement and loading of the arm as harmless and necessary factors for the resolution of the condition. These theories also view shoulder problems from a broader perspective, whereas ‘impingement’ only looks at the biomechanical causes. Today, shoulder problems are seen from a biopsychosocial perspective, where various factors from all three domains can influence pain, function and prognosis.

Although this story goes slightly beyond the purpose of this book, I believe it is important to highlight the issues related to so-called ‘impingement’, as it can be one of the first things that comes up when you google ‘shoulder pain’.

SHOULDER PAIN



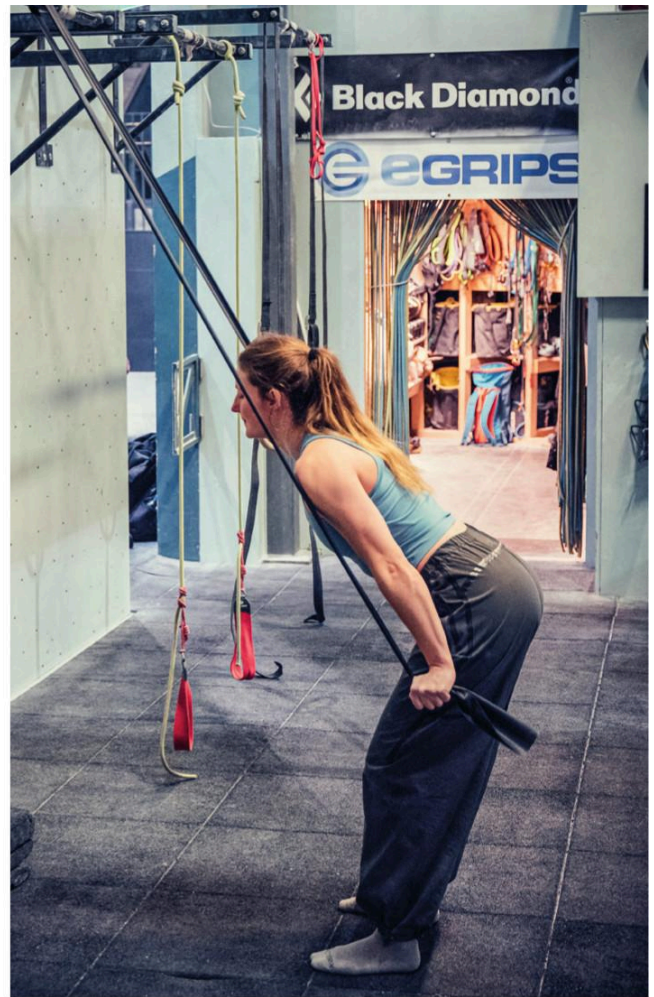
PAINFUL SHOULDERS

Painful shoulders are not, however, exclusively limited to the involvement of the rotator cuff tendons. There are several other structures in the shoulder that can contribute to the experience of shoulder pain, and currently we cannot differentiate these structures from one another in a way that allows us to definitively say, ‘*You have shoulder pain because of [insert structure here]*’. Therefore, rotator cuff tendinopathy falls under the umbrella term subacromial pain syndrome (SAPS), where the name suggests that all structures between the head of the humerus and the roof of the shoulder blade can be involved. This may seem confusing and frustrating, because it’s natural to always want to have a definitive answer as to why we have pain in a certain area. But, on the other hand, it also gives us a great opportunity to tailor treatment. By identifying which movements and positions cause symptoms, what initially triggered the problems, and which exercises, training methods and mental strategies provide the best results for each person, most cases will resolve completely.

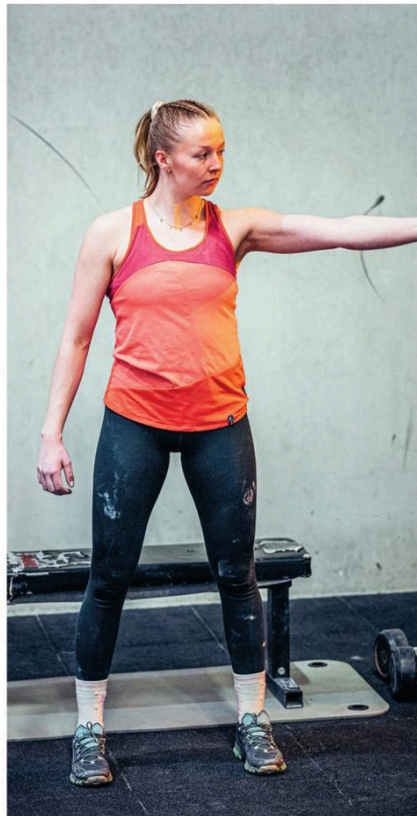
This section has provided an overview of shoulder injuries and problems, ranging from the specific and traumatic to the more load-related and non-specific. Different injuries and different individuals require different approaches, but what is common for all of these injuries and problems is the importance of regaining mobility, stability and strength around the shoulder. Although the shoulder may appear complex and difficult to understand, rehabilitation does not necessarily have to be difficult or complicated. However, it must be tailored to the demands that climbing as a sport places on the shoulders, while also being adapted to your characteristics as a climber.

Here are my go-to exercises for painful shoulders.

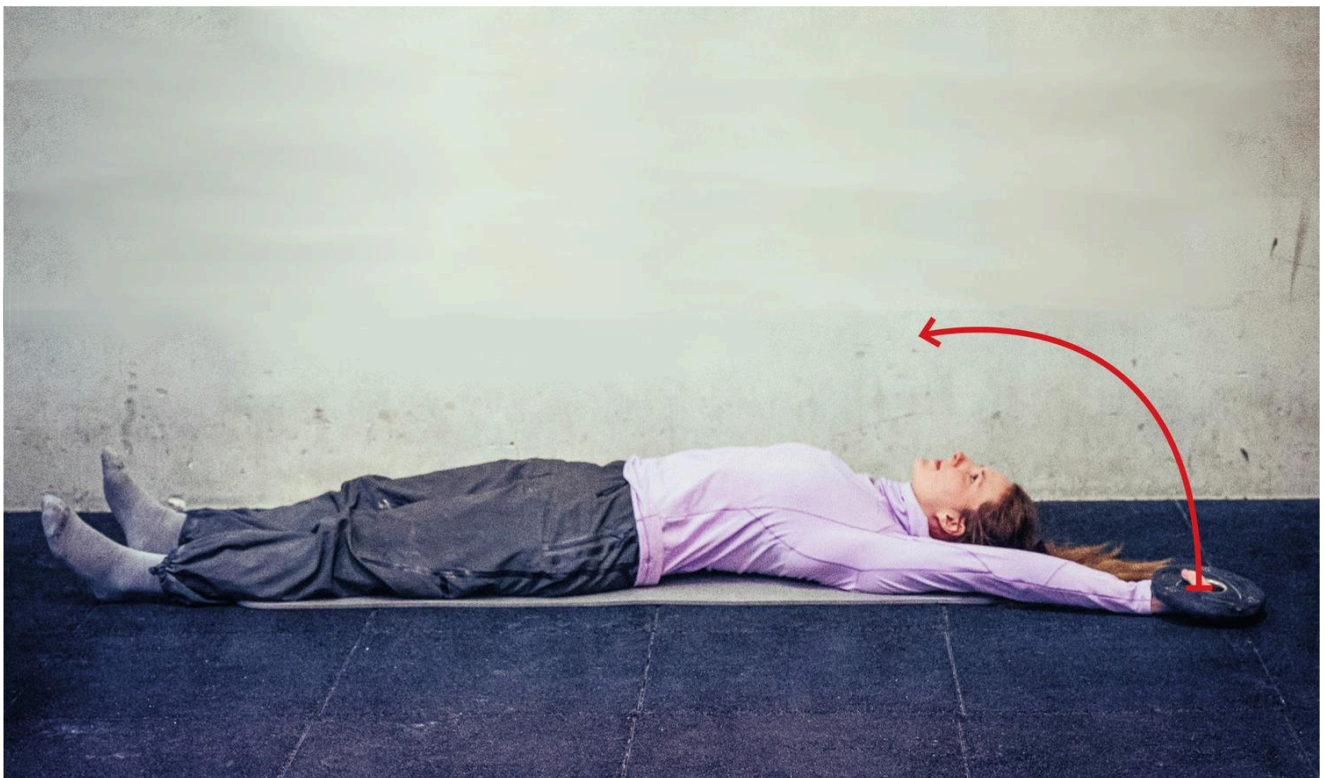
EXERCISES



PULL-DOWNS WITH RESISTANCE BAND. This can be done as a warm-up exercise with about 20 repetitions and 2 sets, the aim of the exercise being to reduce pain when raising the arms above shoulder level.



SCAPTION RAISE. Holding a dumb-bell, raise your arm out at an angle and up over your head. Do 8–10 repetitions and 2–3 sets.



PULL-OVER. Lie on your back and hold a weight disc in one hand directly out in front of you. Bring your arm straight back and down towards the floor. Make sure to keep your back in contact with the floor so that it doesn't arch upwards to help you lower your arm. Note that the bottom position is the most demanding, so adjust the weight so that you have control in this part of the movement. Do 6–8 repetitions and 3 sets as part of your warm-up for climbing, or as part of a strength training programme.



PUSH-UP WITH ROTATION. Do a push-up, then put all your weight on one arm and rotate your torso so that your free arm is pointing up towards the ceiling; repeat with the other side. Do 6–10 repetitions and 3 sets.



MID-RANGE PULL-UP. It's usually the bottom position, and in some cases the top position, that is troublesome. So, the solution for training strength can be to train in the so-called 'mid-range', i.e. avoiding the outer positions, but training heavily in the part of the movement path that is symptom-free.



ASSISTED ONE-ARM LOCK-OFF. Do 2 hangs of 10 seconds at both 90 degrees and 135 degrees of elbow flexion.



PUSH-UP INTO DOWNWARD-FACING DOG. Start lying on your stomach and do a push-up. Then push yourself backwards and bring your head and upper body in between straight arms. Do 6–10 repetitions and 3 sets.

KNEES AND THIGHS

MENISCI, COLLATERAL LIGAMENTS AND CRUCIATE LIGAMENTS

Climbing knee injuries that are not caused by falls are on the rise, and certain climbing positions and movements stand out as triggers: high steps/rockovers, drop knees and heel hooks. What these positions have in common is a bend in the knee with the shin rotated outwards. And the more bending and rotation, the more stress that is placed on the stabilising structures of the knee joint: menisci, collateral ligaments and cruciate ligaments. We usually use these climbing positions and movements when handholds, body positions and moves require us to take the strain off the fingers and relieve the upper body as much as possible. For example, we perform a rockover when the handholds are poor or far apart, moving our centre of gravity to maintain balance or move towards the next grip. The knee is then maximally bent with an externally rotated leg, while we try to straighten the leg to reach for the next hold. A similar knee position can be seen during a drop-knee manoeuvre. Both positions exert immense pressure on the menisci and the medial collateral ligament. A heel hook can be performed at various knee angles, but they typically involve an externally rotated shin with a strong pull from the hamstring muscles to flex the knee. This places a high load on the knee joint overall, but it particularly stresses the menisci and the lateral collateral ligament.

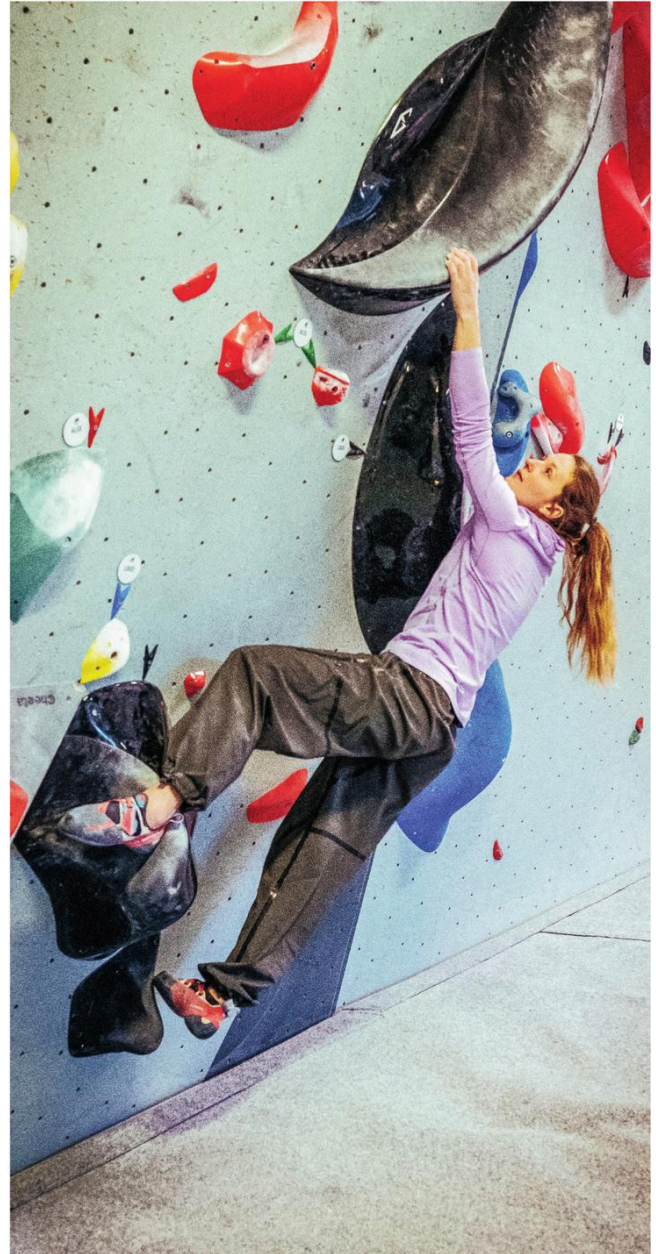
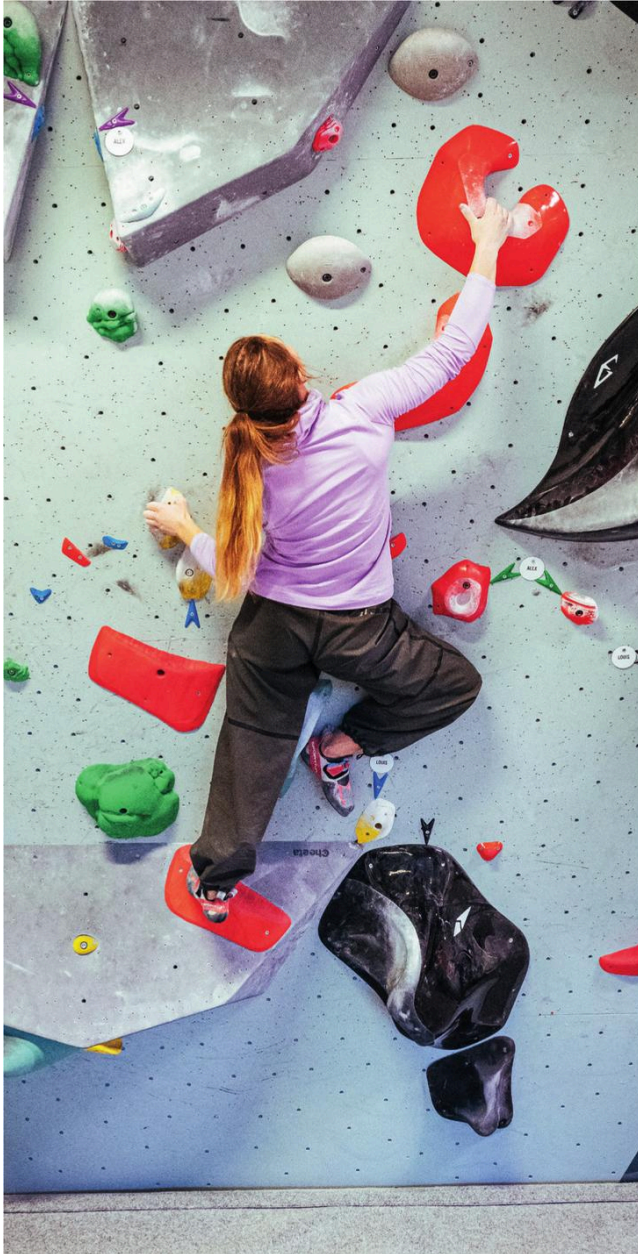
Acute knee injuries usually result in pain and swelling, which limits mobility and function. It's important to clinically examine the knee to assess whether imaging is needed. While ultrasound can detect fluid in the joint and injuries to the collateral ligaments, MRI is necessary for evaluating whether any meniscal or cruciate ligament injuries should be treated surgically. In cases where there is pronounced swelling or locking of the knee joint – i.e. an inability to move it – one should seek medical attention promptly and undergo evaluation for possible surgical treatment.

Can non-fall-related knee injuries be prevented? Well, we can at least reduce the risk by being aware of the positions that push the knee joint to its limits and making a cost-benefit assessment. Some routes and boulders have moves where we simply cannot avoid a high rockover, a deep drop knee, or an awkward heel hook or heel-toe cam. In such cases, we must consider whether we're willing to take the risk. Perhaps we are willing to

risk it on a project, but not during our training sessions. This approach will drastically reduce the number of times the knee is in a vulnerable position, which also reduces the risk of injury. However, we also lose the opportunity to train these positions, which is important for both mastering them technically and being better prepared for encountering them on future projects.

Therefore, my suggestion is to practise rockovers, drop knees and heel hooks as part of your technical training – either in separate sessions or as part of your warm-up exercises. This way, you can use better handholds and footholds to control the load on your knee joint and gradually make it more resilient to higher loads in vulnerable positions.

EXERCISES



Heel hooks at different knee angles and with varying degrees of leg rotation.



COSSACK SQUAT. Start standing with legs apart. Sit down over one foot while keeping the other leg straight. Stand up and repeat on the opposite side. Perform 8–10 repetitions and 2 sets.



BRIDGE AND PULL IN. By placing the heel in a sling and rotating the leg outwards, you can lift your body off the ground and simulate the load in a heel hook. Press the outside of your heel into the sling, lift your body and pull your leg towards you so that your knee is at a 90-degree angle. Perform 6 repetitions and 2 sets.



WALL SQUAT. This is an exercise to simulate high foot movements and rockovers, where you need to rotate your hips and calves and push yourself up from a deep position. Start standing and slowly lower yourself as far down as you can while maintaining contact between the wall and your knees and upper body. Perform 8–10 repetitions and 2 sets.

HAMSTRING INJURIES

‘I went for a heel hook and felt a jerk in the back of my thigh, almost up to my butt. Now it hurts when I walk and straighten my knee. I haven’t even dared to think about heel hooks and climbing.’

The hamstring is a group of three muscles at the back of the thigh. They extend at the hip joint, flex at the knee joint and rotate the shin bone inwards or outwards in relation to the thigh bone, and are susceptible to strains in the whole muscle-tendon-bone unit during heel hooks and the splits.





By pressing the heel against the foothold, the leg muscles contract to either hold the body in one position or pull it towards the next handhold.



In a split, the hamstrings must contract to maintain the extended position.

Hamstring injuries are classified based on their location: either at the knee or at the hip. Injuries in the knee region are generally located to the back and outer side of the knee, where the tendons from the biceps femoris muscle attach to the fibula. Injuries in the hip region are mostly located close the sitting bone, where the majority of the hamstring muscles attach to the pelvis.

While the mechanisms for these two types of injuries have been well documented in running-based sports, they have been explored far less in climbing. We assume that hip-related injuries occur during close-in heel hooks, which involve a high degree of hip flexion. More open hip positions – when the hip isn't bent as much while pulling on a heel and we simultaneously rotate the shin bone outwards for maximum pressure on our heel – predispose us more towards knee-related injuries. These are acute injuries and many climbers will experience a tearing or popping at the back of their thigh.

Strains involving both muscles and tendons should be examined thoroughly to assess the degree and extent of the injury (see table on page [71](#) regarding the grading of injuries). In most cases, the approach to treatment will involve a temporary cessation of hamstring-related movements and a gradual rehabilitation, but with more extensive tendon tears, or a tearing of the tendon attachment itself, surgery may be necessary. Therefore, it is my recommendation that all acute hamstring injuries should be examined by a healthcare professional so that a course of action can be determined as quickly as possible.



In a drop-knee, the hamstring muscles work to help maintain contact with the footholds and to keep the hips close to the wall.

Uzo Ehiogu and colleagues published an article in 2020 which advocated for an alternative management approach for hamstring injuries in climbers compared with hamstring injuries in runners and sprinters. They argued that while running-related sports aim to strengthen the hamstrings through extension – eccentric training – the biomechanics of climbing are the opposite. We must instead strengthen our hamstrings isometrically and concentrically to meet the demands posed by heel hooking. This does not exclude eccentric training of the hamstrings, but emphasises that we must train in a climbing-specific manner.

A rehabilitation plan for hamstring injuries goes through various phases with different goals, representing the different requirements for both the muscle-tendon unit and the entire kinetic chain. In our case as climbers, this involves training the hamstrings in various ways from the time of injury until full recovery, as well as building capacity in the fingers, arms, upper body and core. This latter capacity is often underemphasised, but it's obvious that stronger fingers, arms, upper body and core can give us the ability to relieve the stress on our hamstrings during demanding heel hook positions. The rehabilitation process should therefore focus on becoming a stronger and better climber from a holistic perspective, rather than just training the hamstring muscles in isolation.

MOBILITY EXERCISES

The first stage of rehabilitation after a strain injury is aimed at stretching the muscles.



EXERCISE 1: EXTENDER. Hold your leg behind your knee and straighten your leg as far as you can. Do 12–15 repetitions and 2 sets, daily.



EXERCISE 2: DIVER. Stand on one leg and bring your upper body straight forwards while straightening your back leg. Start with no weight to begin with and then add weight gradually as you can go deeper into the position. Do 6 repetitions and 3 sets every other day.

LOADING EXERCISES

The next stage of rehab aims to load the muscles in different ways.



STRAIGHT HIP EXTENSION. With your leg held out in front of you and with a resistance band around your ankle, move your leg down and back until it is parallel with your standing leg. Do 12 repetitions and 3 sets, twice a week.



LEG CURLS. Lie on your stomach with the resistance band around your ankle, and bend your knee so that your heel moves back towards your body. Do 12 repetitions and 3 sets, twice a week.



SINGLE-LEG BRIDGE. Start lying on your back with one leg on a box, crash pad or similar. Press your heel against the surface and lift your body off the floor. Vary the angle of your knee and how far you turn your leg out to the side. Do 12 repetitions and 3 sets, twice a week.

EXERCISES

HEAVY, SLOW STRENGTH TRAINING



STRAIGHT DEADLIFT. Holding a barbell, drop your upper body forwards while keeping your legs straight. Tighten the backs of your thighs and lift your upper body back to the starting position. Perform for 6 repetitions and 3 sets, twice a week.



DEEP SQUAT. With a barbell on your back, squat down and stand up again. Note that this is a demanding position for the knee joints, so take this into account when determining the weight you use for the exercise. Perform 6 repetitions and 3 sets, twice a week.



BRIDGE AND PULL IN. Press the outside of your heel into the sling, lift your body and pull your leg towards you so that your knee is at a 90-degree angle. Perform 6 repetitions and 3 sets, twice a week.



WEIGHTED SINGLE-LEG BRIDGE. Start lying on your back with one leg on a box, crash pad or similar and place a weight over your hips. Press your heel against the surface and lift your body off the ground. Vary with different knee angles and how much you rotate your leg out to the side. Perform 4–6 repetitions and 3 sets, twice a week.

ANKLE SPRAINS

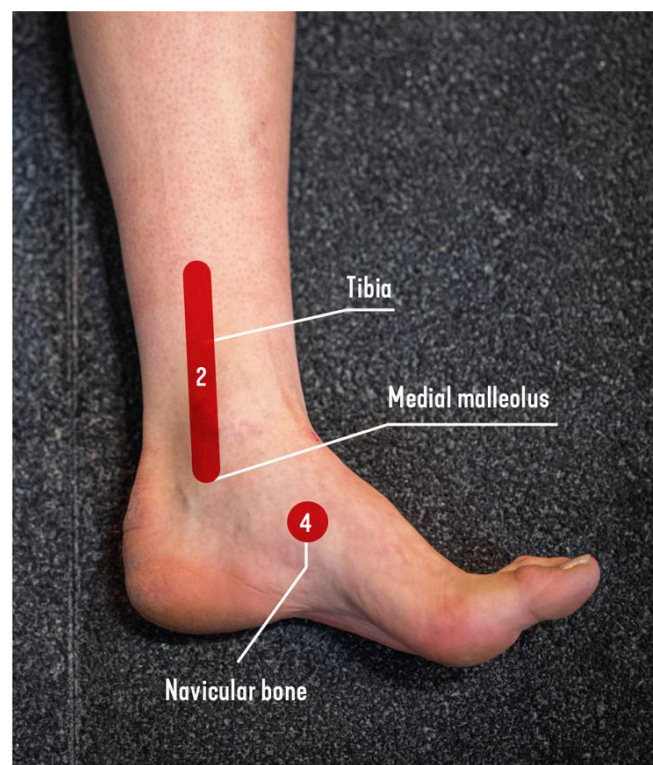
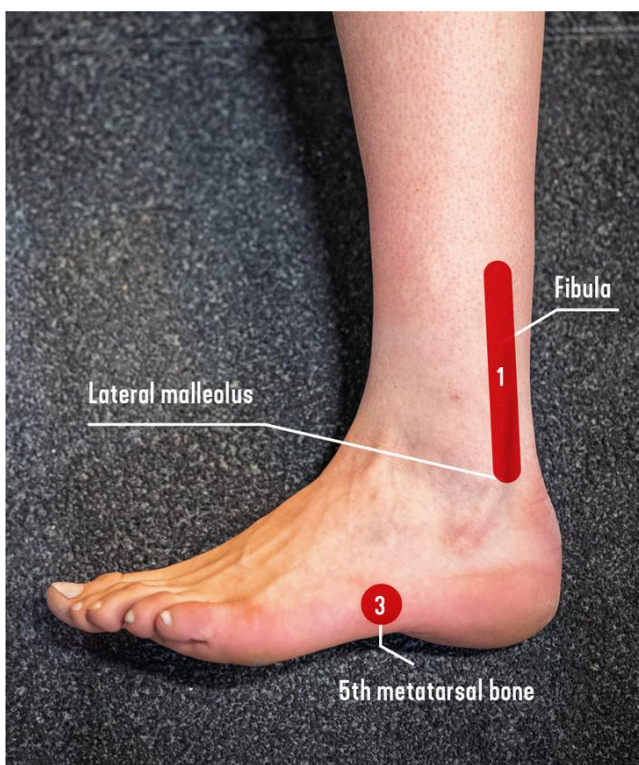
‘I missed the pad a bit and landed wrong. My ankle swelled up almost immediately, and I had to be helped out of the forest and back to the car. It hurts to put weight on my foot, and the ankle feels stiff. What do I do now?’

I’ve had this phone call quite a few times. I’ve also been on the other end of the line, having landed all wrong and sprained my own ankle, so knowing how to handle an ankle sprain is good knowledge to have. As with all acute injuries, you should follow PEACE & LOVE (see pages [18–19](#)), where Protection, Elevation and Compression are most important in the initial phase when you get off the pad and into your car to go home. This means: stop climbing, wrap a compression bandage around your ankle and try to keep your leg elevated above heart level. This lowers the risk of worsening the injury and reduces the swelling around the ankle.

The next step is to rule out a fracture. Sometimes, it’s (unfortunately) easy to detect a fracture, but smaller fractures may not be visible to the naked eye. In such cases, you can use the Ottawa ankle rules (opposite), combined with an assessment of whether or not you can bear full body weight on the injured ankle. If there’s any suspicion of a possible fracture, it’s advisable to visit a hospital for an X-ray so that you’re not walking around on a fracture that needs offloading and proper rest for optimal healing.

If it’s not a fracture, though, the recommendation is to resume weight bearing within the first 48 hours. This doesn’t mean limping around; crutches can be used if needed while moving from point A to point B, but do practise putting some weight on your leg without limping when taking single steps. Since swelling and pain limit mobility in the ankle joint, active exercises should be initiated early in order to increase the range of motion and stimulate muscles. Even if there’s some pain during the exercises, it’s important to pay greater attention to the swelling afterwards: gravity alone will cause the ankle to swell up after you’ve been walking around compared to when it has been elevated above heart level, but the training itself should not lead to an increase in swelling.

A swollen ankle has poorer joint proprioception and thus poorer stability, increasing the risk of re-injury from a premature return to activities that demand high ankle function. Therefore, you should progress slowly from weight-bearing and simple exercises, towards being able to jump and land on the foot and making quick changes in direction before you return to full sporting activity. It’s easy to think that the ankle isn’t so important for us climbers, but anyone who’s sprained their ankle knows how inhibiting it is to climb with a stiff and painful ankle. So, even though it’s quickly possible to start top-roping again, I recommend specifically training an ankle for five minutes a day, five days a week for five weeks after a sprain. This way you’ll regain strength, mobility and stability, and you’ll be able to take falls both on lead and while bouldering, while also regaining confidence in your ankle in all of the movements and positions it is exposed to on the wall.



OTTAWA ANKLE RULES:

1. Bony tenderness along distal 6cm of the posterior edge of fibula or tip of lateral malleolus.
2. Bony tenderness along distal 6cm of the posterior edge of tibia or tip of medial malleolus.
3. Bony tenderness at the base of the 5th metatarsal bone.
4. Bony tenderness at the navicular bone.
5. Inability to bear weight both immediately after injury and for 4 steps during initial examination.

If pain over the ankle bones plus 1, 2 or 5, X-ray of the ankle is recommended.

If pain over the midfoot plus 3, 4 or 5, X-ray of the foot is recommended.

A score of 0 out of 5 indicates less than a 1 per cent probability of fracture.

EXERCISES



HOPPING EXERCISES. Your imagination is the only limit to what you can eventually do! Practising take-offs and landings will help you react faster and stabilise your ankle better in the event of a fall, and it's easy to include such exercises as part of your warm-up for a climbing session. Then you can hop forwards, backwards, sideways, in star formation, and up and down from boxes. Do 6–12 hops per set, and do 3 sets per session.



Wrap a wide resistance band around the inside of your foot and rotate your foot inwards. Do 12–15 repetitions, twice a day.



Wrap a wide resistance band around the outside of your foot and rotate your foot outwards. Do 12–15 repetitions, twice a day.



HEEL RAISE. Stand on the edge of a box, step or similar. Lower your heels past a flat position and then push all the way up on your toes. Do 8–12 repetitions, once a day.

FEET AND TOES

Two sizes down from your regular shoes. Banana-shaped and twisted towards the big toe. A heel cup that tightens over the heel and around the Achilles tendon. All to maximise pressure on the footholds, to keep the heel in place and to allow the toes to function as claws on steep terrain. Climbing shoes are not made for comfort, they are made for function. But is it really that important that they are *SO* tight? And what consequences does this have for your feet and toes?

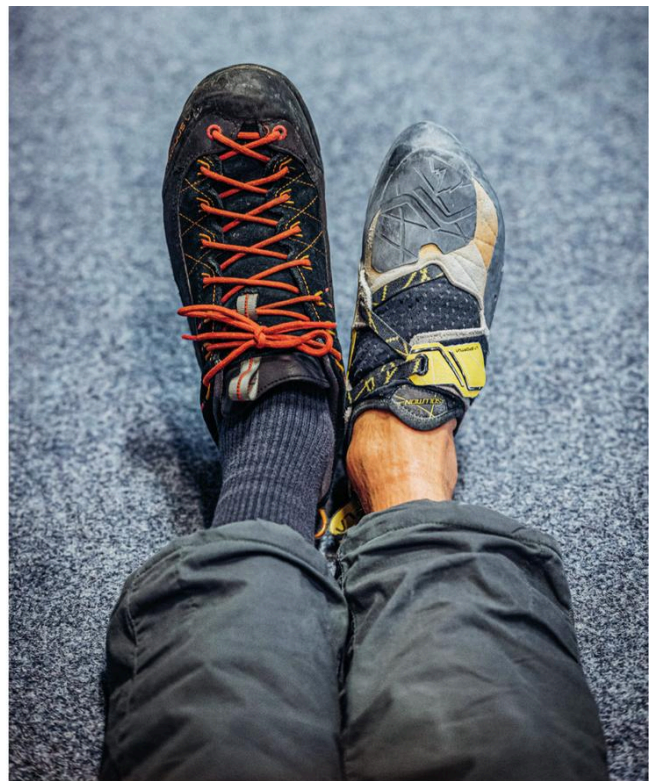
From a personal perspective, it's important for me to have tight shoes. Mine are about 2–2½ sizes smaller than my regular shoes, and I am so used to climbing with tight shoes that it feels more uncomfortable to climb in shoes that are too big. And this is not uncommon among climbers with many years of experience and at higher levels – their climbing shoes are generally a couple of sizes smaller than their regular shoes. Not surprisingly, function is chosen over comfort, but it is important to emphasise that this absolutely does not apply to everyone. The most important thing is to have a climbing shoe that fits well on your foot and allows you to climb without pain. At lower levels, the need for stiffness, precision or feeling isn't as high as in more difficult climbing, and just like I probably don't need an alpine world cup skier's set-up to go skiing, not all climbers need high-end climbing shoes two sizes smaller than their sneakers.

Tight climbing shoes lead most climbers to report discomfort and pain in their feet and toes during climbing. There is a relationship between prolonged use of tight climbing shoes and symptoms such as skin or nail changes, joint pain and bunions, such as *hallux valgus* – a condition where the big toe angles inwards toward the middle of the foot. Since both the extent and severity of such problems seem closely related to climbing level and a reduction in shoe size, it seems obvious that reducing the time spent in very tight shoes would be a good strategy for avoiding such problems. This can be done by using more spacious and comfortable shoes for training, and by taking your shoes off regularly during a climbing session. Your tight and perfectly fitted redpoint shoes shouldn't be worn for long anyway – just long enough for you to send! A conscious approach to choosing shoes for different purposes and the amount of time you wear them for will likely contribute positively to reducing foot and toe problems throughout your climbing life.

However, some conditions may require a more comprehensive approach than just adjusting shoe size, shoe type and duration of use. In my practice, I regularly see inflammation in the base joint of the big toe and the development of bunions. While the former can be treated with cortisone injections into the joint if the inflammation doesn't improve with changes in footwear and load management, managing bunions requires an individualised approach. This ranges from relieving pressure on the big toe through taping, exercises and adaptation of footwear, all the way to surgical treatment. If you have a symptomatic bunion, I recommend contacting a healthcare professional who can help you manage this condition in the best possible way.



X-rays of the author's foot in a climbing shoe. Note the joint angles in the big toe and how all the toes are pressed together to transfer as much force as possible to the tip of the big toe. Thanks to Helene Lund and Unilabs Bryn for the X-rays.



A foot with street shoe size 42 that will fit in a size 39 climbing shoe. It's natural that it's not a comfortable experience and it's not necessary for everyone, all the time. In cases where you really need a tight and precise shoe, make sure you don't wear it for longer than necessary – if you had even considered that option.